

Reproducible Sparse-Concept and Calibration Diagnostics for Knowledge Tracing

A Protocol and Diagnostic Study

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Abstract

Knowledge Tracing (KT) models are typically evaluated with aggregate predictive metrics such as the area under the ROC curve (AUC) and accuracy. These metrics can mask how a model behaves on sparse and cold-start Knowledge Components (KCs)—concepts with limited or no training-fold evidence—even though educational decision support often depends on the reliability of predicted probabilities. Rather than proposing a new model, we treat KT evaluation as a reproducibility and diagnostic problem. We present a diagnostic protocol that combines learner-based and temporal splits, train-only KC-frequency stratification with an explicit strict cold-start group, per-stratum calibration analysis via Expected Calibration Error (ECE), Brier-score decomposition, and reliability diagrams, sample-size-aware reporting, and an operational leakage and predictive-sanity workflow (L1–L8). We instantiate the protocol on ASSISTments 2012, Junyi Academy, and XES3G5M with IRT, DKT, and SimpleKT. Sparse training evidence does not universally degrade discrimination, but it can expose dataset-dependent calibration vulnerability. On ASSISTments 2012, SimpleKT ECE rises from 0.114 on dense KCs to 0.228 on sparse KCs (Limited flag, $N = 415$); other datasets show flat or inverted stratum patterns. On one learner-based fold of that dataset, a global threshold $\tau = 0.7$ yields higher SimpleKT false mastery among advances on sparse than on dense KCs (Limited flag); the SimpleKT ΔFM stays positive on all five training seeds (mean 0.047; four unique partitions). A train-only GKT graph shrinks the seed-42 gap. That calculation is a simulated gate, not a classroom trial. A controlled reduction of training rows for the same originally dense KCs increases within-KC ECE only in some dataset-model cells. We characterize the conditions associated with this vulnerability—sparse mass, evaluation support, and frequency-difficulty coupling—and provide reproducible diagnostics for identifying when it matters. Occupancy reporting remains useful even when the empirical priority of sparse-stratum claims is low. Channel L8 additionally flagged a prediction-label misalignment that aggregate metrics alone would not reveal. We provide an anonymized reproducibility artifact for peer review and will release the public version upon acceptance.

Keywords: Knowledge Tracing; Educational Data Mining; Calibration; Sparse Concepts; Cold-start Knowledge Components; Reproducibility

1 Introduction

Knowledge Tracing (KT) estimates a learner’s evolving mastery state from sequential interactions and predicts future correctness (Corbett and Anderson, 1994; Piech et al., 2015; Abdelrahman et al., 2023). Recent KT models, including memory-based, attention-based, and transformer-style variants, are typically compared using aggregate predictive metrics such as the area under the ROC curve (AUC) and accuracy (ACC) (Zhang et al., 2017; Ghosh et al., 2020; Liu et al., 2023a). These metrics are useful, but they can hide systematic differences on low-frequency knowledge components (KCs), especially when most test interactions are concentrated on dense, frequently observed KCs.

This issue is not merely a reporting detail. In educational decision support, predicted probabilities can be used to decide whether a learner should receive remediation, practice another item, or progress to a new concept. A model that ranks answers well overall may still be poorly calibrated on sparse or limited cold-start KCs. In such cases, high predicted confidence can be misleading even when the model’s aggregate AUC appears competitive. Calibration metrics such as Expected Calibration Error (ECE), Brier score, and reliability diagrams therefore provide a complementary diagnostic view (Guo et al., 2017; Brier, 1950; DeGroot and Fienberg, 1983).

This paper treats KT evaluation itself as a reproducibility and diagnostic problem. We do not propose a new KT architecture, nor is this protocol designed to establish a new state-of-the-art model or to change the “winning” model choice. Its purpose is to diagnose reliability properties—in particular calibration on sparse concepts—that aggregate discrimination metrics may conceal. We specify a dataset-agnostic diagnostic protocol that extends standard KT evaluation with train-only KC-frequency stratification, per-stratum calibration diagnostics, concept-level cold-start analysis, and an operational leakage and predictive-sanity workflow (L1–L8). The framing is intentionally conservative: the goal is reusable evaluation infrastructure for future KT studies—including graph-based and self-supervised approaches whose gains are often claimed on sparse concepts—rather than any claim of state-of-the-art performance. The protocol is not only preventative. In our own instantiation, its predictive-sanity channel flagged a prediction–label misalignment in the temporal evaluation pipeline that aggregate metrics alone would not have revealed (Section 5); we report the audit trail as part of the study.

The scientific question is not whether sparse concepts universally perform worse. Under the conditions of this study, sparse training evidence does not universally degrade discrimination, but it can expose dataset-dependent calibration vulnerability. We characterize the conditions associated with that vulnerability and provide reproducible diagnostics for identifying when it matters. The instantiation supports six bounded findings: (1) lower KC frequency does not imply universal AUC degradation; (2) calibration can be more sensitive than aggregate discrimination in some sparse-concept regimes; (3) the calibration–frequency relationship is dataset-dependent; (4) observable factors such as difficulty, item support, learner exposure, and curriculum position explain part of this heterogeneity; a controlled reduction of training rows for the same originally dense KCs does not universally increase ECE, though SimpleKT on Junyi Academy shows a large within-KC increase; (5) sparse-concept diagnostics are therefore conditionally important rather than equally informative on every KT log; (6) L1–L8 is a reproducible audit workflow, and L8 has empirical value in detecting prediction–label misalignment and the controlled fault classes reported in Appendix J.

1.1 Motivating Diagnostic Gap

Under standard benchmarking protocols, overall KT performance can be misleading because sparse KCs and limited cold-start KCs can exhibit different predictive and calibration behavior from the dense majority. For example, a model may achieve competitive overall AUC on a learner-based split while showing higher calibration error, or unstable AUC estimates, on sparse KC strata—or, on another dataset, no such sparse-stratum degradation at all. This discrepancy matters when predicted probabilities are used to make decisions regarding student remediation, progression, or mastery thresholds. If a model’s probabilities are poorly calibrated on the concepts where evidence is scarcest, educational interventions may be triggered inappropriately, despite a deceptively high population-level discrimination score. The gap is therefore diagnostic rather than architectural: evaluation should report performance and reliability at the KC-frequency level, with occupancy and sample-size context, and with leakage audits that keep pipeline artifacts from being read as scientific findings.

In addressing this gap, this paper makes three contributions:

- **Problem 1 (Hidden Stratum Behavior):** Overall AUC/ACC values can mask heterogeneous predictive behavior or metric instability on low-frequency concepts. **Contribution 1:** We formalize a train-only KC-frequency stratification protocol—with an explicit strict cold-start group ($f_{\text{train}}(c) = 0$) separated from very sparse concepts—that prevents sparse-bucket leakage and enables sample-size-aware KT diagnostics across pre-registered frequency buckets.
- **Problem 2 (Unreliable Probability Estimates):** Aggregate rankings do not reveal whether predicted probabilities correspond to empirical correctness rates, which is crucial for educational decision-making. **Contribution 2:** We evaluate calibration-oriented metrics—ECE, the Brier decomposition, and reliability diagrams—across frequency buckets, and we report when discrimination and calibration diverge. When predicted probabilities are treated as a global mastery/remediation gate, we further simulate stratum-wise false-mastery and miss rates at a locked threshold (Section 4.5). The associated empirical claim is bounded: calibration can be more sensitive than aggregate discrimination in some sparse-concept regimes, the calibration–frequency relationship is dataset-dependent, and a population threshold can then produce a larger false-mastery rate on sparse KCs than on dense KCs for SimpleKT on ASSISTments 2012 (positive ΔFM on all five training seeds, four unique partitions; Limited occupancy).
- **Problem 3 (Hidden Evaluation Pipeline Contamination):** KT evaluation pipelines can suffer from subtle, undocumented data leakage and alignment errors across folds or splits. **Contribution 3:** We operationalize eight named leakage and predictive-sanity checks into a reproducible KT evaluation workflow with verifiable artifacts. Channels L1–L7 instantiate standard evaluation safeguards (split hygiene, train-only definitions, validation-only selection) as inspectable checks; they are not offered as a new auditing methodology. Channel L8 is a warm-cohort predictive-sanity test that, in this study, flagged a temporal prediction–label misalignment (Appendix I). A subsequent controlled fault-injection validation (Appendix J) shows that the same signals are sensitive to some tested alignment-fault classes (label-identity shifts; global row-index mismatch) and insensitive to others (adjacent-timestep prediction shifts that remain autocorrelated). We do not claim that L8 detects all evaluation bugs.

Our experimental investigation is structured around four Research Questions (RQs):

- **RQ1 (Stratum-level Predictive Heterogeneity):** How does predictive performance vary across KC-frequency strata, and is lower training frequency associated with systematic AUC degradation across datasets?
- **RQ2 (Calibration Profiles by Frequency):** How do model calibration profiles change across different concept frequency strata, and is calibration more sensitive than aggregate discrimination in some sparse-concept regimes?
- **RQ3 (Explanatory Conditions):** Under what dataset and concept conditions is lower training frequency associated with meaningful calibration degradation, and when are sparse-concept diagnostics informative rather than uniformly required?
- **RQ4 (Limited Cold-start Concept Diagnostics):** How do baseline KT models behave on KCs with zero or limited training-fold frequency under temporal splits?

RQ1 exercises Contribution 1, RQ2 exercises Contribution 2 (including the single-fold GKT/CL4KT instantiation and simulated gate), RQ3 uses Contributions 1–2 together with covariate-adjusted associations and a controlled sparsification experiment (Appendix G), and RQ4 draws jointly on Contributions 1 and 3. The remainder of this paper is organized as follows. Section 2 reviews KT evaluation, sparse-concept analysis, and calibration. Section 3 presents the diagnostic protocol. Section 4 reports the experimental design and diagnostic results. Section 5 discusses implications and limitations.

2 Related Work

2.1 Knowledge Tracing Models and Benchmarking

Knowledge Tracing models dynamic learner proficiency by leveraging historical interaction sequences to forecast future responses (Abdelrahman et al., 2023). Classical approaches, such as Item Response Theory (IRT) (Rasch, 1960; Embretson and Reise, 2000) and Bayesian Knowledge Tracing (BKT) (Corbett and Anderson, 1994), model static ability and latent mastery states, respectively. Deep Knowledge Tracing (DKT) (Piech et al., 2015) shifted the field toward recurrent neural networks, followed by memory-augmented architectures such as DKVMN (Zhang et al., 2017) and attention-based models such as AKT (Ghosh et al., 2020) and SimpleKT (Liu et al., 2023a). Benchmarking toolkits such as pyKT (Liu et al., 2022) have substantially improved the standardization of preprocessing and model evaluation, and have documented how easily evaluation choices can distort reported performance. Our work is complementary to this benchmarking line: rather than standardizing model training and comparison, we standardize diagnostic reporting—how predictive and probabilistic quality should be broken down, audited, and interpreted once predictions have been produced. We therefore use representative baselines from this literature to instantiate the protocol, not to compete with them.

2.2 Graph-Augmented Knowledge Tracing

Contemporary architectures frequently leverage graph neural networks to capture structural dependencies—such as prerequisites, semantic similarities, or co-occurrences—between knowledge components (KCs) (Abdelrahman et al., 2023). GKT (Nakagawa et al., 2019) and GIKT (Yang et al., 2021) construct KC–item or KC–KC dependency graphs, while contrastive and

self-supervised approaches such as CL4KT (Lee et al., 2022) learn from structurally augmented views. These architectures capture rich relational signal, but they also enlarge the evaluation attack surface: graph construction, augmentation, and self-supervised objectives introduce additional channels through which information can leak across splits, and their intended gains are typically claimed precisely on low-frequency concepts, where evaluation is least stable. Reliable stratum-level and leakage-audited evaluation is therefore a prerequisite for credible claims in this family. The main tables instantiate the protocol on sequential baselines. Section 4.5 additionally applies the same train-only strata to GKT and to a CL4KT protocol adapter on one ASSISTments 2012 fold, as a diagnostic check rather than a state-of-the-art comparison.

2.3 Sparse and Cold-start Knowledge Components

Educational datasets exhibit long-tail distributions in which a small set of dense KCs dominates the interaction volume while many KCs carry sparse training evidence (Choi et al., 2020; Liu et al., 2023b). Sparse and cold-start conditions in KT should be distinguished from the cold-start scenarios typically studied in recommender systems: recommender systems usually define cold-start at the user or item level (new users or items)—a challenge also studied in KT as the new student problem (Bhattacharjee and Wayllace, 2025)—whereas the variant of the problem addressed here concerns concepts with insufficient interaction history in the training data (Pelánek, 2015b; Choffin et al., 2019). Our protocol operationalizes this distinction through train-only KC-frequency stratification that separates strict cold-start concepts ($f_{\text{train}}(c) = 0$) from very sparse, sparse, medium, and dense strata, so that concepts with no training evidence are never conflated with concepts that have limited but nonzero evidence.

The term sparse has also been used in a different, model-internal sense. sparseKT (Huang et al., 2023) introduces k -sparse attention that selects the most relevant historical interactions to improve robustness and generalization. This is orthogonal to our usage: sparseKT sparsifies attention over interaction histories, whereas our protocol diagnoses low-frequency and zero-frequency KCs via train-only counts. Sparse attention and sparse-concept evaluation should not be conflated.

Recent work addresses cold-start KT from complementary angles. csKT (Bai et al., 2025) targets short interaction sequences via kernel bias and cone attention; CLST (Jung et al., 2025) reformulates KT with natural-language descriptions of exercises and KCs; and further LLM-based approaches include attribute-aware KT (Guo et al., 2024) and option-weighted KT (Kim et al., 2024). These methods target learner-level, sequence-level, system-level, or semantic cold-start, whereas our protocol defines and audits cold-start at the concept level through train-only frequency. Because this paper proposes a diagnostic protocol rather than a new architecture or attention mechanism, we do not compare against these model-level methods; instead, the proposed diagnostics can be applied to them—sparseKT-like and cold-start KT models alike—in future evaluations.

2.4 Calibration of Probabilistic Predictions

Calibration is essential for KT because predicted probabilities of correctness can directly drive instructional interventions. In student modeling, Pelánek’s evaluation framework is the central prior reference for probability-level assessment using the Brier score and calibration curves (Pelánek, 2015a), establishing that student models should be evaluated beyond binary accuracy. In the broader machine-learning literature, deep networks can achieve high

discrimination while remaining poorly calibrated and overconfident (Guo et al., 2017); calibration error is commonly quantified via the Expected Calibration Error (ECE) (Naeini et al., 2015), and the Brier score (Brier, 1950) admits a decomposition into uncertainty, reliability, and resolution (DeGroot and Fienberg, 1983) that separates intrinsic task difficulty from model miscalibration.

Our protocol builds directly on this tradition, and differs from it in how the calibration evidence is organized: prior calibration-oriented student-model evaluation is typically reported at the population level, rather than around train-only KC-frequency strata, an explicit concept-level strict cold-start group, and sample-size-aware bucket diagnostics with reliability flags. We package those calibration diagnostics with an operational leakage and predictive-sanity workflow; we do not claim that channels L1–L7 constitute a new auditing theory. Complementary to our focus on evaluation diagnostics, recent work has begun to explore methodological corrections for per-item bias and stranded discrimination (Yan et al., 2026); our protocol provides the diagnostic foundation to assess whether such corrections effectively improve reliability across varying frequency strata.

2.5 Reproducibility, Leakage Control, and Statistical Testing

As sequence modeling architectures grow increasingly complex, ensuring rigorous reproducibility and strict leakage control has become critical for reliable evaluation (Kapoor and Narayanan, 2023; Gervet et al., 2020; Kim and Kim, 2026). The field faces a leakage and reproducibility crisis, and KT is particularly susceptible to silent evaluation errors due to its complex sequence structure. To mitigate this, we operationalize L1–L8 as named channels tied to verifiable artifacts: L1–L7 encode standard safeguards for KT pipelines, and L8 adds a warm-cohort predictive-sanity check. For comparing competitive models, DeLong’s test provides a standard approach for correlated ROC curves (DeLong et al., 1988). Statistical testing is used only where prediction-level outputs permit paired comparison; otherwise, we report descriptive diagnostics with sample-size-aware interpretation.

3 Diagnostic Protocol

This section specifies the protocol in dataset-agnostic terms; its instantiation on three datasets follows in Section 4.

3.1 Data Schema

Each dataset is normalized into a common interaction schema

$$(u_t, i_t, c_t, r_t, \tau_t), \tag{1}$$

where u_t is the learner, i_t the item, c_t the knowledge component (KC), $r_t \in \{0, 1\}$ the correctness, and τ_t the timestamp. Multi-skill items are represented by a one-row-per-KC convention: each interaction involving multiple KCs is pre-expanded into one row per KC by the dataset source, so that each processed row corresponds to exactly one (learner, item, KC, correctness, timestamp) tuple. Specifically, XES3G5M is loaded from the `kc_level` split provided by the dataset authors (Liu et al., 2023b), which already stores one KC per row; ASSISTments 2012 (ASSISTmentsData, 2012) is read from the standard release file in which each row carries a single `skill_id`; Junyi Academy (Junyi Academy, 2019) items

are inherently single-skill. Our `preprocess.py` pipeline performs no additional expansion. KC-frequency stratification (Section 3.3) is computed on these pre-expanded rows using training-fold counts only.

3.2 Split Modes and Cold-start Groups

The protocol evaluates under three split modes, which answer different diagnostic questions and must not be conflated. The learner-based split assigns learners disjointly to train/validation/test partitions and tests generalization to unseen learners. The temporal split trains on earlier interactions and evaluates on later ones, testing future-oriented forecasting. The limited cold-start concept analysis evaluates KCs by their training-fold frequency under the temporal split, using four groups: strict cold-start KCs ($f_{\text{train}}(c) = 0$), the limited cold-start groups **k5** ($f_{\text{train}}(c) \leq 5$) and **k10** ($f_{\text{train}}(c) \leq 10$), and the complementary warm group. The strict group isolates concepts with no training evidence; **k5** and **k10** probe near-zero evidence regimes; the warm group serves as the within-split sanity reference used by audit channel L8 (Section 3.6).

3.3 KC-frequency Stratification

For each training fold, we compute the training frequency of a KC c as

$$f_{\text{train}}(c) = \sum_{(u,i,c',r,\tau) \in D_{\text{train}}} \mathbb{I}[c' = c]. \quad (2)$$

KCs are assigned to five pre-registered groups:

$$\text{Strict Cold-start: } f_{\text{train}}(c) = 0, \quad (3)$$

$$\text{Very Sparse: } 0 < f_{\text{train}}(c) < 20, \quad (4)$$

$$\text{Sparse: } 20 \leq f_{\text{train}}(c) < 100, \quad (5)$$

$$\text{Medium: } 100 \leq f_{\text{train}}(c) < 500, \quad (6)$$

$$\text{Dense: } f_{\text{train}}(c) \geq 500. \quad (7)$$

The assignment must use only training-fold counts. Validation and test frequencies may be logged for auditing but must not determine the bucket. It is important to acknowledge a conceptual caveat regarding this definition: sparsity is not an absolute property of the underlying knowledge itself. Rather, it is partially an artifact of human authorship in the Q-Matrix construction. A concept might appear sparse not because the fundamental knowledge is rarely practiced, but because domain experts defined it at an overly granular level or tagged it inconsistently across items. Consequently, while our strata reliably isolate low-frequency training patterns, they reflect the operational sparsity of the dataset’s specific KC mapping rather than an ontological truth about the learning domain. This operational sparsity is additionally conditioned on the preprocessing pipeline: filtering of malformed or below-minimum-length sequences and dataset-specific multi-skill handling alter per-KC training-fold counts, so stratum membership should be read as a property of the processed dataset under our pipeline, not of the raw logs.

3.4 Calibration Metrics

Given predictions p_i and labels y_i , ECE with $M = 15$ bins is

$$\text{ECE} = \sum_m 1^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (8)$$

where $\text{acc}(B_m)$ is empirical correctness and $\text{conf}(B_m)$ is average predicted confidence in bin B_m .

The Brier score is

$$\text{Brier} = \frac{1}{N} \sum_i (p_i - y_i)^2. \quad (9)$$

We use the standard decomposition

$$\text{Brier} = \text{UNC} - \text{RES} + \text{REL}, \quad (10)$$

where the three components are defined as follows. Let $\bar{y}$ denote the global correctness rate, $M = 15$ the number of calibration bins, n_m the number of samples in bin m , acc_m the empirical accuracy in bin m , conf_m the average predicted confidence in bin m , and N the total number of samples.

$$\text{UNC} = \bar{y}(1 - \bar{y}). \quad (11)$$

$$\text{REL} = \frac{1}{N} \sum_{m=1}^M n_m (\text{conf}_m - \text{acc}_m)^2. \quad (12)$$

$$\text{RES} = \frac{1}{N} \sum_{m=1}^M n_m (\text{acc}_m - \bar{y})^2. \quad (13)$$

UNC measures the intrinsic uncertainty of the prediction task and is independent of the model; REL measures the deviation between predicted probabilities and observed frequencies, where lower values indicate better reliability; RES measures how well the model separates events with different outcome rates from the global mean, where higher values indicate better resolution. Note that because this formulation bins continuous probabilities into M discrete bins, the resulting decomposition is an empirical approximation and components need not sum exactly to the directly computed Brier score. Calibration is reported both overall and per KC stratum.

Discrimination and calibration answer different questions. AUC measures whether a model ranks correct responses above incorrect ones, whereas calibration measures whether predicted probabilities correspond to empirical correctness rates. The distinction matters operationally because probability thresholds are often used to trigger remediation, suggest additional practice, or license concept advancement: a model with high AUC can still exhibit substantial miscalibration and thereby induce inappropriate instructional interventions. The converse also holds—a model can appear well calibrated while being uninformative—which is why the protocol requires calibration to be interpreted jointly with discrimination and resolution, never in isolation.

3.5 Sample-size-aware Reliability Flags

To ensure methodological rigor, we introduce sample-size-aware reliability flags that reflect the stability of the metric estimates:

- **R (Reliable):** $N \geq 1000$
- **L (Limited):** $100 \leq N < 1000$
- **I (Insufficient):** $N < 100$

These flags provide descriptive context about the stability of the evaluation metrics and do not certify whether sufficient training evidence was provided to the model. Metrics for Insufficient strata ($N < 100$) are reported descriptively but should not be used as stable evidence of model behavior.

3.6 Eight-channel Leakage & Predictive Sanity Audit

Table 1 summarizes the workflow. We do not propose L1–L7 as a new auditing methodology: they operationalize standard evaluation safeguards—disjoint splits, train-only transforms and group definitions, and validation-only selection—as named KT checks with inspectable artifacts. Their value is reproducibility (every reported table can be paired with a PASS/FAIL log), not theoretical novelty. Channel L8 is different in kind: it is a predictive-sanity test that compares warm-cohort temporal AUC against a static IRT reference, so that near-random deep-KT performance on well-evidenced concepts is treated as a pipeline warning rather than as a scientific finding. In this instantiation L8 detected a real defect (Appendix I); a later fault-injection validation measures sensitivity on additional alignment-fault classes (Appendix J).

Table 1: L1–L8 as a reproducible audit workflow, not a new auditing theory. L1–L7 operationalize standard evaluation safeguards for KT artifacts. L8 is the only channel with an empirical defect-detection case study (Appendix I) and a controlled fault-injection validation (Appendix J).

Ch.	Standard safeguard	KT-specific operationalization	Empirical demonstration
L1	Disjoint units; no future-to-past leakage	User-overlap and timestamp-order checks on interaction sequences (<code>split_audit.csv</code>)	Preventative. PASS on the reported splits.
L2	Fit transforms on training data only	No global normalization of the full log before splitting (<code>preprocess.py</code>)	Preventative. PASS.
L3	No test-informed feature maps	Static dataset KC / Q-matrix mapping (<code>kc_map.json</code>)	Preventative. PASS.
L4	Do not define evaluation subgroups on the test set	Train-only f_{train} buckets, including an explicit strict cold-start group (<code>kc_strata.csv</code>)	Preventative. PASS. KT-specific encoding of a standard rule.
L5	No test-set metric tuning	ECE / Brier computed post-hoc from exported prediction files (<code>baseline_runner.py</code>)	Preventative. PASS.
L6	Validation-only model selection	Early stopping and selection on the validation fold only (<code>baseline_runner.py</code>)	Preventative. PASS.
L7	Group labels independent of test outcomes	Strict / k5 / k10 / warm defined from training-fold frequency only (<code>split_constructor.py</code>)	Preventative. PASS. KT-specific encoding of a standard rule.
L8	(Not a generic split-hygiene item)	Warm-cohort temporal AUC versus a static IRT reference, plus association and identity-consistency signals; near-random deep-KT AUC on well-evidenced KCs is a pipeline warning (<code>temporal_audit.py</code> ; <code>a7_l8_fault_injection.py</code>)	Empirical case + validation. Flagged a prediction–label misalignment (Appendix I). Sensitive to injected label-identity shifts and row-index mismatch; not to adjacent prediction shifts (Appendix J).

3.7 Diagnostic Pipeline

Figure 1 illustrates the pipeline.

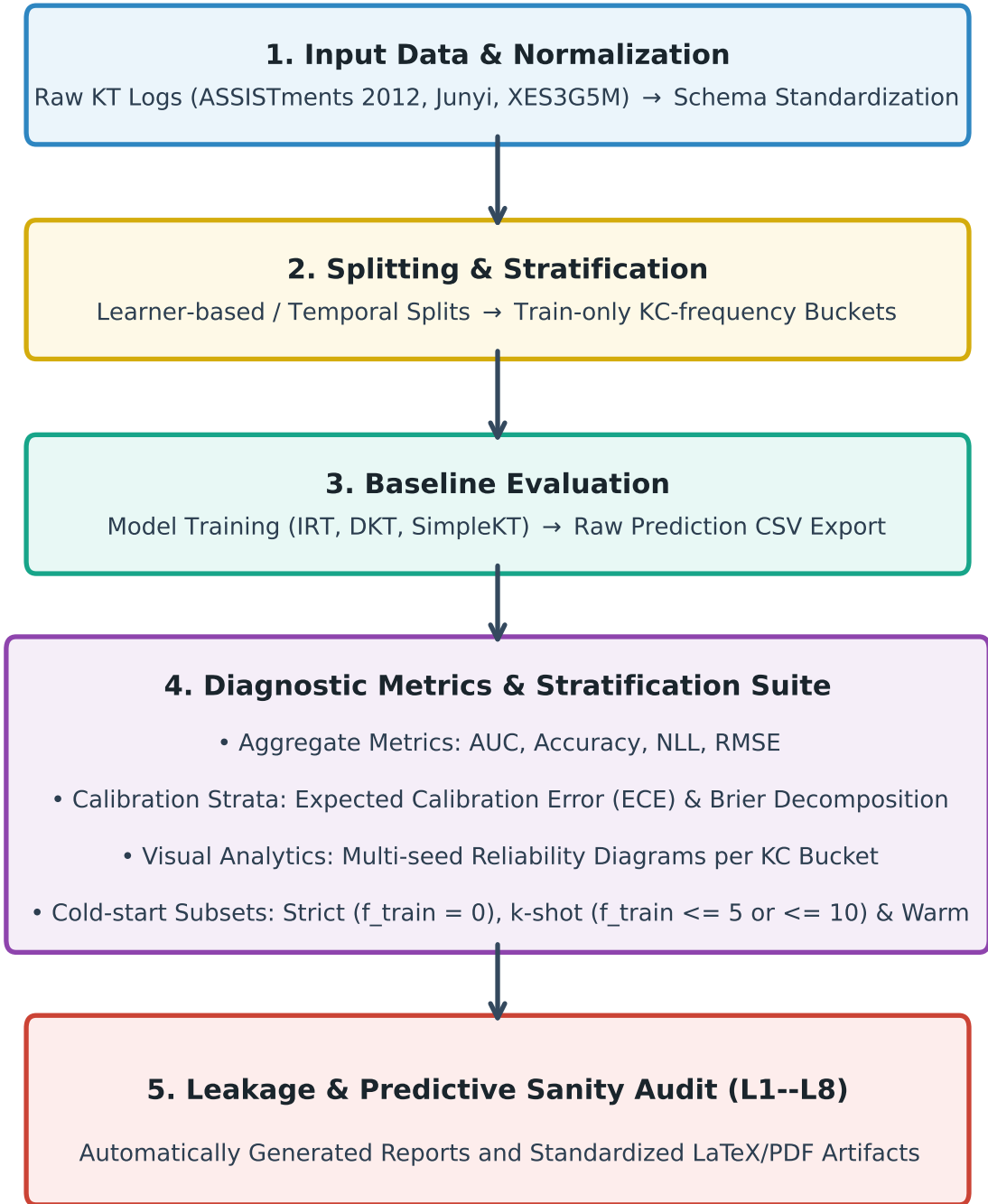


Figure 1: Overview of the reproducible sparse-concept and calibration diagnostic pipeline for Knowledge Tracing. (Note: In the figure, the term “k-shot” denotes “limited” cold-start KCs where $f_{\text{train}} \leq 5$ or ≤ 10 , as discussed in the text.)

3.8 Design Principles

To ensure the mathematical validity and operational reproducibility of the diagnostics, our protocol is governed by four core design principles:

- **P1. Train-only Definitions:** All knowledge component (KC) buckets and limited cold-start groups must be defined using training-fold information only. This prevents sparse-bucket leakage.

- **P2. Prediction-level Export:** Rather than computing downstream diagnostics in-memory during baseline training, our protocol mandates exporting raw prediction probabilities and ground-truth labels directly to disk. All metrics are computed post-hoc from these standardized artifacts.
- **P3. Sample-size-aware Interpretation:** Bucket-level metrics must never be reported in isolation. They must be accompanied by the exact number of KCs and interaction events, contextualized by the Reliability Flags (R, L, I), to differentiate between true performance degradation and metric instability.
- **P4. Leakage-audited Reporting:** No diagnostic result is considered complete unless accompanied by an explicit, verifiable check. This is operationalized as the L1–L8 reproducible audit workflow (an operational checklist with predictive-sanity validation), not as a new auditing theory.

3.9 Explanatory Analyses

After prediction-level exports are written, two additional analyses speak to RQ3 without changing stratum definitions or official training files. First, a KC-level confounding analysis (Section 5.3) relates ECE to training-only covariates (frequency, a difficulty proxy, item support, learner exposure, and a curriculum-position proxy). These associations are observational. Second, a controlled sparsification experiment (Appendix G) downsamples training rows of originally dense KCs while holding the test set, labels, KC identifiers, and all other KCs fixed, then retrains DKT and SimpleKT. That experiment tests whether reducing training count for the *same* KC is followed by a within-KC ECE increase; it does not identify a causal law for real-world sparsity.

4 Experiments and Diagnostic Analysis

As mapped in Section 1, our experimental investigation is structured around four Research Questions (RQs): RQ1 exercises Contribution 1 (stratum-level heterogeneity), RQ2 exercises Contribution 2 (calibration profiles), including a single-fold instantiation of GKT and a CL4KT adapter plus a simulated global-threshold decision analysis on ASSISTments 2012, RQ3 states the dataset conditions under which sparse-concept calibration diagnostics are informative and reports covariate-adjusted plus controlled-sparsification evidence, and RQ4 draws jointly on Contributions 1 and 3.

4.1 Experimental Setup and Datasets

We evaluate our protocol on three diverse, public, de-identified benchmark educational datasets: ASSISTments 2012 (assist2012) (ASSISTmentsData, 2012; Liu et al., 2022), Junyi Academy (junyi) (Junyi Academy, 2019), and XES3G5M (xes3g5m) (Liu et al., 2023b). No personally identifiable learner information is processed in this study. The preprocessing pipeline (Section 3) retains only well-formed interaction records; Table 2 presents the resulting cohort statistics.

Dataset provenance. ASSISTments 2012 was obtained from the ASSISTments 2012–2013 public release with affect predictions, as documented by ASSISTmentsData and the pyKT dataset documentation (ASSISTmentsData, 2012; Liu et al., 2022). The Junyi Academy data used in this study corresponds to the Junyi Academy Online Learning Activity Dataset

released on Kaggle (Junyi Academy, 2019), as reflected by the post-preprocessing cohort size in Table 2 (71,014 learners, 25,784 items, 1,326 KCs, and 16.2M interactions); it should not be confused with the earlier Junyi2015 release, which has different learner/item/KC counts. XES3G5M was obtained from the benchmark release of Liu et al. (2023b). All counts in Table 2 are post-preprocessing counts produced by our normalization pipeline. Because these are processed rather than raw counts, they reflect dataset- and pipeline-specific steps that can shift totals in either direction relative to a raw interaction log: filtering of malformed records and short sequences reduces counts, whereas datasets distributed in a one-row-per-KC (KC-level) form, such as XES3G5M, contain more rows than a one-row-per-interaction count would suggest. Because the KC-frequency strata (Section 3.3) are computed on these processed training-fold counts, stratum membership is conditional on this pipeline. Consequently, the number of KCs appearing in bucket-level temporal tables can be smaller than the dataset-level #KCs in Table 2 when some KCs have no valid evaluated prediction rows after temporal filtering and sequence construction; we report these evaluated-KC totals explicitly below Table 11.

Learner-based splits serve as the main predictive evaluation setting (reported in Tables 3 and 5, with detailed calibration results in Appendix C, Table 12), because they test generalization to unseen learners while preserving enough interactions for stable bucket-level analysis. Temporal splits serve as a complementary stress test for future-oriented and cold-start diagnostics (reported in Table 9 and Appendix B). Keeping the two modes separate prevents learner-level and curriculum-level forecasting challenges from being conflated.

For all deep baselines on learner-based splits, we report mean $\pm$ standard deviation over four unique student partitions. The processed partitions for seeds 2025 and 2026 are identical (`fold_2 = fold_3` on all three datasets); those two runs therefore vary training initialization on a shared student split. We average their metrics into one partition-level value before summarizing Tables 3, 5, and 12. Frequency buckets use each fold’s training-only f_{train} . The simulated-gate robustness table still lists all five training seeds without dropping one after the fact (Table 8). For temporal splits, after the label-alignment correction documented in Appendix I, we report deterministic metrics from the single corrected evaluation run (seed 42) used to regenerate Figure 3 and the temporal calibration tables. This single-run design is intentional but limited: unlike learner-based cross-validation, which randomly partitions students, the temporal split relies on a single deterministic chronological boundary (an 80/20 time cutoff) to simulate future forecasting. Because the partition is fixed, the reported temporal results constitute a fixed-cutoff stress test; they do not estimate variance across temporal cut-points or neural initializations. We do not claim that initialization variance is absent, and we return to this limitation in Section 5. IRT is deterministic given a fold, and its learner-based variation reflects fold differences only.

Table 2: Dataset and Splitting Cohort Statistics

Dataset	Split Mode	#Learners	#Items	#KCs	#Interactions	#Train	#Valid	#Test	Avg. Seq. Len.
ASSISTments 2012	Learner-based	27,806	52,672	265	2,657,490	1,865,217	258,123	534,150	95.57
ASSISTments 2012	Temporal	27,806	52,672	265	2,657,490	1,860,242	265,749	531,499	95.57
Junyi Academy	Learner-based	71,014	25,784	1,326	16,215,567	11,373,990	1,572,555	3,269,022	228.34
Junyi Academy	Temporal	71,014	25,784	1,326	16,215,567	11,350,896	1,621,556	3,243,115	228.34
XES3G5M	Learner-based	18,066	7,653	866	7,953,709	5,572,247	792,317	1,589,145	440.26
XES3G5M	Temporal	18,066	7,653	866	7,953,709	5,567,596	795,370	1,590,743	440.26

Figure 2 illustrates heterogeneous KC-frequency distributions across datasets. Although dense KCs dominate the interaction volume, sparse and very sparse KCs remain important diagnostic strata because they expose low-frequency regions where aggregate metrics may be less informative.

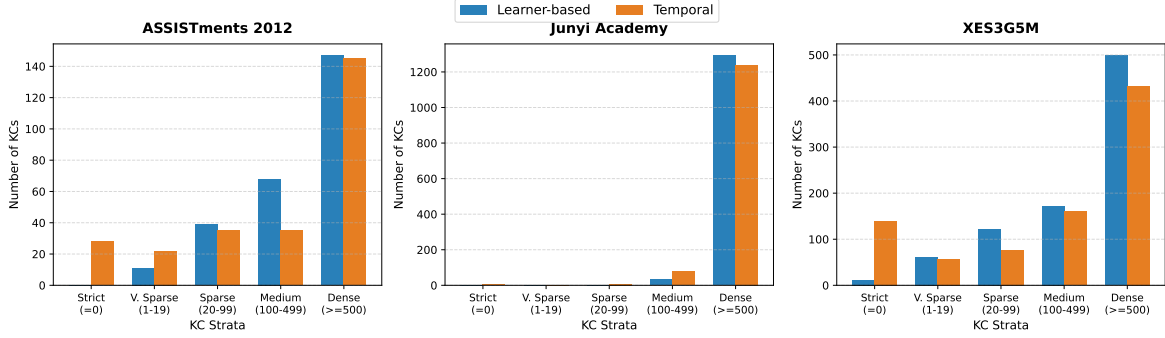


Figure 2: KC-frequency strata distribution across ASSISTments 2012, Junyi Academy, and XES3G5M. **Strict Cold-start:** $f_{\text{train}} = 0$; **Very Sparse:** $0 < f_{\text{train}} < 20$; **Sparse:** $20 \leq f_{\text{train}} < 100$; **Medium:** $100 \leq f_{\text{train}} < 500$; **Dense:** $f_{\text{train}} \geq 500$. Strict cold-start KCs are plotted separately from very sparse KCs to avoid conflating zero-frequency concepts with low-frequency concepts.

4.2 Baseline Implementations

We initially evaluated BKT but replaced it with IRT after observing numerical instability in Expectation-Maximization (EM)-based `pyBKT` estimation; details are provided in Appendix H. We evaluate three representative classes of KT models:

1. **Item Response Theory (IRT 1PL):** a classical logistic baseline (one-parameter logistic Rasch model) estimating a static learner ability and concept difficulty from training interactions. Because the model relies on learner identities, unseen learners in learner-based splits have no estimated ability parameters; in our implementation the prediction then falls back to a global, base-rate-like probability. Under learner-based splits we therefore treat IRT as a base-rate reference rather than as a discriminative baseline ($\text{AUC} = 0.50$ by construction). Under temporal splits, where learners are observed during training, IRT functions as a genuine classical baseline and serves as the static reference for audit channel L8 (Appendix I).
2. **Deep Knowledge Tracing (DKT):** The pioneering recurrent neural network KT architecture mapping student sequential states via Long Short-Term Memory (LSTM) or Recurrent Neural Network (RNN) cells.
3. **SimpleKT:** A modern, self-attention-based model utilizing key-query similarities between concept embeddings to track mastery.

All deep learning models are trained with validation-based early stopping to prevent hyperparameter leakage.

4.3 RQ1: Stratum-level Predictive Heterogeneity

To address RQ1, we first inspect the overall performance of all three baselines in Table 3. We report AUC, ACC, the negative log-likelihood (NLL), and the root mean squared error (RMSE). SimpleKT and DKT achieve competitive aggregate AUC scores under learner-based splits, with DKT slightly leading SimpleKT on ASSISTments 2012 (0.6980 ± 0.0013 vs. 0.6840 ± 0.0025). As discussed, IRT performs at random ($\text{AUC} = 0.5000$) on learner-based splits due to its base-rate-like prediction for unseen learners. Consequently, discriminative comparison under the learner-based split effectively contrasts the two deep baselines,

Table 3: Overall Performance under Learner-based Split

Dataset	Split	Model	AUC	ACC	NLL	RMSE
ASSISTments 2012	Learner-based	IRT	0.5000	0.6973 ± 0.0004	0.6131 ± 0.0003	0.4594 ± 0.0002
	Learner-based	DKT	0.6979 ± 0.0014	0.7182 ± 0.0014	0.5836 ± 0.0049	0.4400 ± 0.0012
	Learner-based	SimpleKT	0.6837 ± 0.0025	0.6996 ± 0.0032	0.7013 ± 0.0055	0.4603 ± 0.0020
Junyi Academy	Learner-based	IRT	0.5000	0.7053 ± 0.0018	0.6063 ± 0.0015	0.4559 ± 0.0008
	Learner-based	DKT	0.7320 ± 0.0009	0.7343 ± 0.0015	0.5440 ± 0.0021	0.4248 ± 0.0009
	Learner-based	SimpleKT	0.7231 ± 0.0030	0.7274 ± 0.0015	0.6070 ± 0.0145	0.4340 ± 0.0024
XES3G5M	Learner-based	IRT	0.5000	0.7961 ± 0.0031	0.5625 ± 0.0016	0.4315 ± 0.0009
	Learner-based	DKT	0.8171 ± 0.0022	0.8327 ± 0.0032	0.3932 ± 0.0083	0.3495 ± 0.0032
	Learner-based	SimpleKT	0.7557 ± 0.0013	0.8067 ± 0.0037	0.6202 ± 0.0152	0.3930 ± 0.0037

Note: Mean $\pm$ standard deviation over four unique learner-based partitions. Seeds 2025 and 2026 share one student split (`fold_2=fold_3`); their metrics are averaged into one partition value before the four-partition summary. IRT is the classical baseline. Under learner-based splits, IRT’s AUC remains at 0.5000 because unseen learners do not have estimated ability parameters, causing the model to fall back to a constant or base-rate-like prediction. Consequently, its ACC mainly reflects majority-class base-rate and thresholding behavior rather than discriminative ranking ability.

with IRT serving only as a base-rate anchor; the temporal split (Section 4.7 and Appendix B), where IRT becomes a genuine classical baseline, complements this setting, and the need for a classical baseline that exploits item difficulty for unseen learners is flagged as a limitation in Section 5.3. Under temporal splits (Table 10 in Appendix B), where learners are observed, IRT’s AUC rises to 0.5913 on ASSISTments 2012, while SimpleKT (0.6731) slightly outperforms DKT (0.6606).

Table 4: DeLong Tests for Overall AUC, DKT vs. SimpleKT (Learner-based Split)

Dataset	DKT vs SimpleKT
ASSISTments 2012	$p = 7.10 \times 10^{-82*}$
Junyi Academy	$p = 8.07 \times 10^{-144*}$
XES3G5M	$p < 0.001^*$

Note: One test per dataset; Bonferroni-corrected significance threshold $\alpha = 0.05/3 \approx 0.0167$. * indicates significance after correction. DeLong tests are computed from paired prediction-level outputs aligned on stable test-instance identities via an inner join, so that only instances present in both models’ exported predictions are compared; rows appearing in only one model, due to model-specific sequence filtering, are discarded before the test statistic is computed. Comparisons against the IRT reference are not reported: because IRT produces a constant, base-rate-like prediction for unseen learners under learner-based splits (Section 4.2), DeLong tests against it are degenerate.

Table 4 reports paired DeLong tests for the overall learner-based AUC comparison between DKT and SimpleKT.

All three comparisons are statistically significant after Bonferroni correction, confirming that aggregate predictive quality differs between the two deep baselines.

The stratum-level breakdown in Table 5 serves as descriptive diagnostic evidence of heterogeneous stratum behavior. Predictive performance varies by dataset and stratum rather than following a monotonic AUC degradation. ASSISTments 2012 exhibits partial non-monotonic changes; Junyi Academy, under the learner-based split, does not activate the sparse or very sparse strata at the pre-registered thresholds; and XES3G5M presents an inverted counter-pattern (higher AUC on sparse than on dense KCs for both deep baselines). Lower KC frequency therefore does not imply universal AUC degradation (Finding 1). Aggregate metrics can hide this dataset-specific stratum-level behavior; they do not always hide a sparse-AUC failure, because that failure is not present in every dataset.

The evaluated-KC totals are reported in the note below Table 5 to clarify that bucket-level analyses cover only KCs with valid evaluated prediction rows. Furthermore, the sum of evaluated events across all strata is strictly smaller than the raw test partition sizes reported in Table 2. This data attrition (e.g., 0.8% in ASSISTments 2012, 1.1% in Junyi Academy,

Table 5: Knowledge Tracing Performance Breakdown by KC-frequency Strata (Learner-based Split)

Dataset	Model	Bucket	Rel.	#KCs	#Events	AUC	ACC	NLL	RMSE
ASSISTments 2012	IRT	dense	R	146	523,971	0.5000	0.6993 $\pm$ 0.0007	0.6115 $\pm$ 0.0006	0.4586 $\pm$ 0.0003
		medium	R	68	5,963	0.5000	0.5363 $\pm$ 0.0190	0.7466 $\pm$ 0.0158	0.5236 $\pm$ 0.0071
		sparse	L	38	415	0.5000	0.6184 $\pm$ 0.0555	0.6786 $\pm$ 0.0460	0.4916 $\pm$ 0.0221
		Very Sparse	I	6	16	0.5000	0.4717 $\pm$ 0.0666	0.8002 $\pm$ 0.0553	0.5470 $\pm$ 0.0237
	DKT	Strict Cold-start	I	1	4	-	0.0000	1.1915	0.6962
		dense	R	146	523,971	0.6968 $\pm$ 0.0014	0.7185 $\pm$ 0.0014	0.5806 $\pm$ 0.0040	0.4394 $\pm$ 0.0011
		medium	R	68	5,963	0.7574 $\pm$ 0.0141	0.6897 $\pm$ 0.0162	0.8037 $\pm$ 0.0867	0.4790 $\pm$ 0.0151
		sparse	L	38	415	0.7287 $\pm$ 0.0106	0.6978 $\pm$ 0.0224	1.1568 $\pm$ 0.0687	0.5001 $\pm$ 0.0099
	SimpleKT	Very Sparse	I	6	16	0.9369 $\pm$ 0.0530	0.8184 $\pm$ 0.0685	0.5974 $\pm$ 0.3958	0.3665 $\pm$ 0.1221
		Strict Cold-start	I	1	4	-	0.5000	0.7013	0.5017
		dense	R	146	523,971	0.6822 $\pm$ 0.0024	0.6997 $\pm$ 0.0033	0.6998 $\pm$ 0.0053	0.4600 $\pm$ 0.0021
		medium	R	68	5,963	0.7546 $\pm$ 0.0153	0.6912 $\pm$ 0.0130	0.8028 $\pm$ 0.0313	0.4766 $\pm$ 0.0090
		sparse	L	38	415	0.7167 $\pm$ 0.0225	0.6962 $\pm$ 0.0108	1.1446 $\pm$ 0.0853	0.5024 $\pm$ 0.0115
		Very Sparse	I	6	16	0.8271 $\pm$ 0.0503	0.7610 $\pm$ 0.0536	0.9108 $\pm$ 0.1743	0.4633 $\pm$ 0.0498
		Strict Cold-start	I	1	4	-	0.3750	0.7471	0.5252
Junyi Academy	IRT	dense	R	1,295	3,226,541	0.5000	0.7055 $\pm$ 0.0018	0.6062 $\pm$ 0.0015	0.4558 $\pm$ 0.0008
		medium	R	31	3,706	0.5000	0.5820 $\pm$ 0.0083	0.7125 $\pm$ 0.0071	0.5078 $\pm$ 0.0033
	DKT	dense	R	1,295	3,226,541	0.7322 $\pm$ 0.0010	0.7345 $\pm$ 0.0015	0.5434 $\pm$ 0.0022	0.4247 $\pm$ 0.0009
		medium	R	31	3,706	0.6054 $\pm$ 0.0299	0.5839 $\pm$ 0.0172	1.0719 $\pm$ 0.0693	0.5546 $\pm$ 0.0129
	SimpleKT	dense	R	1,295	3,232,614	0.7231 $\pm$ 0.0030	0.7275 $\pm$ 0.0015	0.6069 $\pm$ 0.0145	0.4340 $\pm$ 0.0024
		medium	R	31	3,836	0.6865 $\pm$ 0.0241	0.6442 $\pm$ 0.0191	0.6866 $\pm$ 0.0291	0.4803 $\pm$ 0.0106
XES3G5M	IRT	dense	R	498	1,269,345	0.5000	0.7962 $\pm$ 0.0031	0.5624 $\pm$ 0.0016	0.4314 $\pm$ 0.0009
		medium	R	173	12,889	0.5000	0.7838 $\pm$ 0.0057	0.5697 $\pm$ 0.0031	0.4355 $\pm$ 0.0018
		sparse	R	120	1,969	0.5000	0.7536 $\pm$ 0.0093	0.5873 $\pm$ 0.0053	0.4452 $\pm$ 0.0029
		Very Sparse	L	38	114	0.5000	0.7266 $\pm$ 0.0418	0.6030 $\pm$ 0.0243	0.4536 $\pm$ 0.0132
	DKT	Strict Cold-start	I	4	5	0.5000	0.6528 $\pm$ 0.2953	0.6462 $\pm$ 0.1720	0.4698 $\pm$ 0.0907
		dense	R	498	1,268,696	0.8169 $\pm$ 0.0020	0.8326 $\pm$ 0.0031	0.3916 $\pm$ 0.0073	0.3494 $\pm$ 0.0031
		medium	R	173	12,980	0.8578 $\pm$ 0.0067	0.8429 $\pm$ 0.0091	0.4973 $\pm$ 0.0926	0.3506 $\pm$ 0.0140
		sparse	R	120	2,010	0.8571 $\pm$ 0.0041	0.8286 $\pm$ 0.0070	0.7222 $\pm$ 0.1461	0.3756 $\pm$ 0.0115
	SimpleKT	Very Sparse	L	39	112	0.8381 $\pm$ 0.0479	0.7882 $\pm$ 0.0405	1.1497 $\pm$ 0.5610	0.4228 $\pm$ 0.0435
		Strict Cold-start	I	3	4	0.5536 $\pm$ 0.2778	0.4028 $\pm$ 0.3081	0.7240 $\pm$ 0.0792	0.5141 $\pm$ 0.0379
		dense	R	498	1,268,696	0.7546 $\pm$ 0.0013	0.8064 $\pm$ 0.0036	0.6200 $\pm$ 0.0150	0.3932 $\pm$ 0.0037
		medium	R	173	12,980	0.8338 $\pm$ 0.0057	0.8368 $\pm$ 0.0066	0.6179 $\pm$ 0.0388	0.3674 $\pm$ 0.0082
		sparse	R	120	2,010	0.8468 $\pm$ 0.0023	0.8180 $\pm$ 0.0037	0.6965 $\pm$ 0.0463	0.3842 $\pm$ 0.0065
		Very Sparse	L	39	112	0.8406 $\pm$ 0.0238	0.7958 $\pm$ 0.0176	0.9461 $\pm$ 0.0604	0.4152 $\pm$ 0.0134
		Strict Cold-start	I	3	4	0.4286 $\pm$ 0.1010	0.3194 $\pm$ 0.4016	1.0953 $\pm$ 0.5143	0.6294 $\pm$ 0.1740

Note: Reliability flags are assigned by the number of evaluated test events: Reliable (R: $N \geq 1000$), Limited (L: $100 \leq N < 1000$), and Insufficient (I: $N < 100$). Results in Insufficient buckets are descriptive only. Mean $\pm$ standard deviation and event counts are over four unique learner-based partitions (seeds 2025 and 2026 averaged first). After test-fold filtering and KC-strata matching, the evaluated learner-based KC totals are 259/265 for ASSISTments 2012, 1,326/1,326 for Junyi Academy, and 833/866 for XES3G5M. Within a stratum, evaluated event counts may differ across models. This bidirectional variance occurs because (1) sequence models often drop initial interactions lacking prior state or apply minimum-length filtering, reducing dense-event totals compared to classical IRT; (2) IRT implementations often fail to output predictions for concepts with insufficient training history, while deep models still generate predictions via initialized embeddings, increasing their sparse-event totals; and (3) codebase-specific windowing and padding choices cause further discrepancies.

and 19.2% in XES3G5M) strongly correlates with average sequence length (Table 2). It is produced by sequence-windowing (dropped prefixes and short-sequence filters), not by dropping KCs according to training frequency, so it should not be read as a second frequency filter.

We note that the behavior observed in sparse and very sparse strata represents a combination of true performance differences and metric instability. Therefore, bucket-level diagnostics must be interpreted jointly with sample size, as formalized by our reliability flags. For example, DKT achieves an apparently excellent AUC of 0.9045 ± 0.0667 on the ASSISTments 2012 very sparse stratum; however, because this evaluates only 19 test events across 7 KCs, it triggers an Insufficient (I) flag, indicating the metric is purely descriptive and highly unstable.

An instructive counter-pattern emerges on XES3G5M. Even after separating strict cold-start KCs from the very sparse stratum, DKT and SimpleKT show higher AUC on sparse KCs than on dense KCs (DKT: 0.8571 vs. 0.8169; SimpleKT: 0.8468 vs. 0.7546), and the very sparse stratum remains competitive for both (DKT: 0.8381; SimpleKT: 0.8406), subject to its Limited flag. Strict cold-start rows are reported for completeness only. Several non-exclusive explanations are plausible:

- XES3G5M’s hierarchical KC tree may place sparse KCs among advanced concepts reached only by learners who have mastered prerequisites, creating a ceiling effect.
- Item-level features may carry more signal than KC-level features, partially decoupling KC frequency from predictive difficulty.
- Multi-skill expansion may concentrate predictions on items whose primary skill is well trained even when the auxiliary skill is sparse.

Disambiguating these explanations requires item-level error analysis beyond the scope of this protocol paper. However, the counter-pattern itself reinforces the central point: frequency is not a universal predictor of AUC degradation, and bucket-level diagnostics reveal heterogeneous phenomena that overall AUC cannot.

4.4 RQ2: Calibration Profiles by Frequency

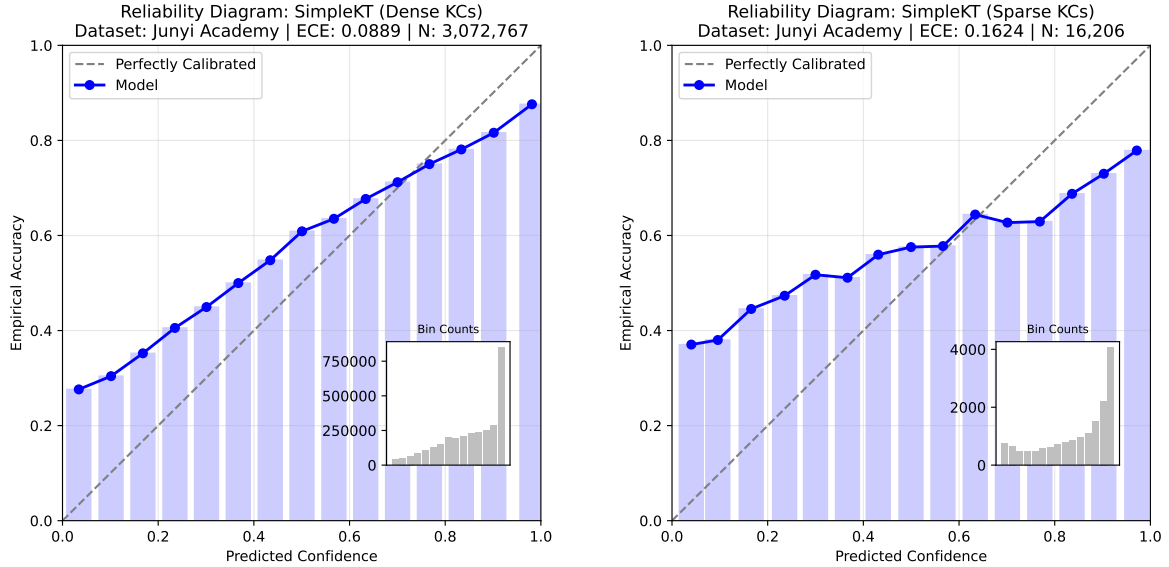
To answer RQ2, we analyze Expected Calibration Error (ECE) and Brier Score decomposition across frequency strata. On Junyi Academy under the learner-based split, the sparse and very sparse buckets are empty because all KCs have training-fold frequency at least 100. This absence is itself a diagnostic observation: Junyi does not exhibit learner-based sparse-KC concerns under the pre-registered thresholds.

Detailed calibration results are reported in Appendix C, Table 12.

Notably, IRT exhibits very low ECE on several dense strata (e.g., 0.0033 on ASSISTments 2012) yet has zero Brier resolution ($RES = 0$) across strata and random discrimination ($AUC = 0.50$) on learner-based splits. This occurs because IRT lacks ability estimates for unseen learners and predicts close to the global correctness rate, yielding low reliability error ($REL \approx 0$) but no discriminative power. This illustrates a critical point: a model can appear well-calibrated precisely by being uninformative.

Calibration gradient on ASSISTments 2012. Under our experimental conditions, ASSISTments 2012 provides the clearest example of a sparse-stratum calibration gradient. For SimpleKT, ECE increases from 0.1136 ± 0.0066 (dense) to 0.1541 (medium) and 0.2280 ± 0.0197 (sparse), and reaches 0.2451 ± 0.0511 in the very sparse stratum; the sparse and very sparse figures carry Limited and Insufficient flags, respectively, and are interpreted accordingly. A similar monotonic pattern appears in DKT’s reliability component, with REL rising from 0.0053 (dense) to 0.0301 (medium) and 0.0624 (sparse). On the same dense stratum, IRT reports far lower ECE (0.0031) than DKT (0.0602) and SimpleKT (0.1136), but per the base-rate analysis above this does not indicate superior predictive reliability: IRT has $RES = 0$ and $AUC = 0.50$ there. Because aggregate AUC remains competitive while sparse ECE rises on this dataset, calibration is a more sensitive diagnostic dimension than aggregate discrimination *in this regime* (Finding 2). The same ECE rise is not observed on every dataset (Finding 3; RQ3).

Because temporal splits stress future-concept generalization, we use the Junyi temporal split in Figure 3 to visualize calibration behavior. The dense stratum has a lower ECE (0.0889), whereas the sparse stratum shows higher ECE (0.1624) while retaining meaningful discrimination ($AUC = 0.6529$). This comparison highlights why calibration and discrimination must be interpreted jointly.



(a) Reliability diagram of SimpleKT on Junyi temporal split for Dense KCs.

(b) Reliability diagram of SimpleKT on Junyi temporal split for Sparse KCs.

Figure 3: Reliability diagrams of SimpleKT on the Junyi temporal split after the label-alignment correction (Appendix I). The dashed diagonal denotes perfect calibration; source values are in Table 13. The sparse stratum has higher calibration error than the dense stratum while retaining meaningful discrimination ($\text{AUC} = 0.6529$), illustrating that calibration and discrimination should be interpreted jointly.

4.5 Relational Instantiation and Simulated Threshold Decisions

Finding 2 is a calibration gradient on sequential baselines. Two follow-up questions are whether that gradient is model-specific on the same split, and whether it changes a *thresholded* decision rather than only ECE. Both analyses use the locked ASSISTments 2012 learner-based fold corresponding to seed 42—the same prediction export as the DKT and SimpleKT files used for the GKT comparison—not the four-partition means in Tables 3 and 12. Sparse occupancy on this fold is $N = 444$ (Limited), versus $N = 415$ in the four-partition calibration table; very-sparse $N = 11$ remains Insufficient and is not used for claims.

GKT and a CL4KT adapter. GKT (Nakagawa et al., 2019) is the pyKT implementation with a *train-only* consecutive-KC transition graph (265 KCs; 15,193 nonzero edges). We do not use the pyKT helper that concatenates train and test CSVs before counting transitions. CL4KT is a protocol adapter: a causal transformer trained with the mask/crop/permute/replace views of Lee et al. (2022) on *training* sequences only. It is not a byte-identical official checkpoint, so we do not compare published CL4KT leaderboard AUC.

Table 6 does not overturn sequential models as a leaderboard exercise. Sparse AUC remains higher for GKT (0.783) and the CL4KT adapter (0.761) than for DKT (0.726). What the protocol changes is the *calibration* reading of Finding 2: on this seed-42 fold, SimpleKT sparse ECE is 0.210 and DKT 0.234, whereas train-only GKT sparse ECE is 0.098 and the adapter 0.119 (four-partition SimpleKT sparse ECE in Table 12 is 0.2280 ± 0.0197). The dense-to-sparse ECE rise on ASSISTments is therefore not a property of the dataset alone. We do not claim that GKT or CL4KT should replace SimpleKT in production, and we do not run GIKT or additional datasets.

A full-log GKT graph adds 1,050 edges that never appear in the train-only adjacency.

Early-stopping valid AUC is essentially identical (0.730 vs. 0.730), but test AUC rises from 0.716 to 0.730. Graph leakage of this form can therefore look honest on the validation curve and still move the reported test number. All GKT figures in Tables 6 and 7 use the train-only graph.

Table 6: Same ASSISTments 2012 learner-based fold (seed 42) as the threshold simulation. Metrics are a single run, not the four-partition means in Tables 3 and 12. GKT uses a train-only transition graph. CL4KT is a protocol adapter (Lee et al. contrastive views on *training* sequences only), not a byte-identical official checkpoint. Sparse $N = 444$ is Limited. Very-sparse $N = 11$ is omitted (Insufficient).

Model	AUC (all)	ECE (all)	AUC (sparse)	ECE (sparse)
DKT	0.699	0.060	0.726	0.234
SimpleKT	0.684	0.118	0.745	0.210
GKT (train-only)	0.716	0.037	0.783	0.098
CL4KT (adapter)	0.721	0.009	0.761	0.119

Note: A full-log GKT graph (pyKT default: train+test transitions) adds 1,050 edges that are absent from the train-only graph. Valid AUC is essentially unchanged (0.730 vs. 0.730); test AUC rises from 0.716 to 0.730. Reported GKT numbers use the train-only graph.

Global τ , stratum-wise error. C4 in Table 16 remains qualitative intended use; we do not infer from public logs whether a system actually thresholds p . When that use is *assumed*, the same global gate can be simulated on existing probabilities. Advance (skip remediation) if $p \geq \tau$; otherwise trigger remediation. The locked grid is $\tau \in \{0.5, 0.6, 0.7, 0.8\}$; τ is never chosen on sparse events. The display threshold is $\tau = 0.7$. Co-primary outcomes are false mastery among advances, $FM = P(y = 0 \mid p \geq \tau)$, and missed failures, $Miss = P(p \geq \tau \mid y = 0)$. $E[FM] = E[1 - p \mid p \geq \tau]$ is the advance error a calibrated model would expect.

Table 7 reports $\tau = 0.7$ on the seed-42 fold, where GKT and the CL4KT adapter can be compared. For SimpleKT, FM is 0.196 on dense KCs and 0.268 on sparse KCs ($\Delta FM = +0.072$). Miss does *not* rise on sparse ($\Delta Miss = -0.032$). DKT shows a similar seed-42 FM gap (+0.096). Train-only GKT shrinks ΔFM to +0.015 (sparse FM 0.220 vs. dense 0.205). The CL4KT adapter is intermediate ($\Delta FM = +0.055$). Sparse KCs are also harder on this fold (mean correctness 0.556 vs. 0.700), so a prevalence difference is expected. Calibration excess $FM - E[FM]$ remains large for SimpleKT sparse (0.268 vs. expected 0.050) and smaller for GKT sparse (0.220 vs. 0.157). The locked τ grid does not reverse the SimpleKT ΔFM sign on this fold. IRT 1PL never advances at $\tau \geq 0.7$ here ($\bar{p} \approx 0.696$) and is omitted from the table.

Across the five learner-based *training seeds*, SimpleKT ΔFM at $\tau = 0.7$ stays positive on every seed (mean 0.047, sd 0.033; Table 8). Per-seed sparse N lies in $\{376, \dots, 444\}$ (mean 413, Limited), compared with the four-partition mean $N = 415$ in Table 12; both are Limited, and the join is train-only f_{train} on the prediction export. A KC-clustered bootstrap on seed 42 gives a 95% interval $[0.006, 0.138]$ for that seed’s ΔFM —excluding 0, but wide, as occupancy requires. The 2025/2026 pair shares a split; both of those ΔFM values are still positive (0.010 and 0.020). DKT ΔFM is positive on only three of five seeds and is not treated as a five-run gate finding. GKT and CL4KT are not repeated across seeds.

Junyi Academy has an empty learner-based sparse bucket on all five seeds, so the same gate cannot be scored there. On XES3G5M, SimpleKT ECE is flat (Finding 3), sparse N is Reliable on every seed (mean 2,000 on the prediction export; Table 12 reports 2,010), and ΔFM at $\tau = 0.7$ is negative on all five seeds (mean -0.018), but $\Delta Miss$ is positive on all five (mean $+0.112$): among actual incorrect answers, the system still advances more often on

sparse than on dense KCs. A flat ECE therefore does not imply a flat miss rate. Sparse XES KCs are easier than dense ones on seed 42 (mean correctness 0.754 vs. 0.645), consistent with the inverted discrimination pattern in Table 5. We pre-registered that both absolute seed-42 gaps would be smaller than on ASSISTments SimpleKT; that joint test failed, and we do not retune it.

This simulation is a decision-error diagnostic, not a mastery policy, an RCT, or a post-hoc calibrator.

Table 7: Simulated global-threshold decision error at $\tau = 0.7$ on ASSISTments 2012, learner-based fold 0 (seed 42). Advance if $p \geq \tau$. $FM = P(y = 0 \mid p \geq \tau)$; $E[FM] = E[1 - p \mid p \geq \tau]$ is the error rate a calibrated model would expect among advances; Miss = $P(p \geq \tau \mid y = 0)$; UR = $P(p < \tau \mid y = 1)$. Sparse $N = 444$ is Limited (n_{advance} is smaller). GKT uses the train-only graph. CL4KT is a protocol adapter. This is not an instructional trial.

Model	Stratum	Flag	N	FM	$E[FM]$	Miss	UR
SimpleKT	dense	R	528,018	0.196	0.113	0.352	0.381
SimpleKT	sparse	L	444	0.268	0.050	0.320	0.304
DKT	dense	R	528,018	0.200	0.147	0.398	0.316
DKT	sparse	L	444	0.296	0.057	0.365	0.308
GKT (train-only)	dense	R	528,018	0.205	0.163	0.455	0.244
GKT (train-only)	sparse	L	444	0.220	0.157	0.234	0.340
CL4KT (adapter)	dense	R	528,018	0.185	0.175	0.359	0.322
CL4KT (adapter)	sparse	L	444	0.240	0.116	0.244	0.385

Note: Sparse mean correctness is 0.556 versus 0.700 on dense. Excess FM = $FM - E[FM]$ is 0.218 for SimpleKT sparse versus 0.063 for GKT sparse. The locked τ grid $\{0.5, 0.6, 0.7, 0.8\}$ is reported in the artifact; τ is never selected on sparse events.

Table 8: Five-run robustness of the simulated gate at $\tau = 0.7$ on ASSISTments 2012 (seeds 42, 2024, 2025, 2026, 2027). Seeds 2025 and 2026 share one learner partition; unique splits are four. $\Delta FM = FM_{\text{sparse}} - FM_{\text{dense}}$. Sparse occupancy is Limited on every seed (mean $N = 413$ on the prediction export; Table 12 reports four-partition mean $N = 415$). GKT and the CL4KT adapter remain a single-seed check (Table 7).

Model	Mean ΔFM	SD	Seeds $\Delta FM > 0$	Mean $\Delta Miss$	SD	Mean sparse N
SimpleKT	0.047	0.033	5/5 seeds	0.012	0.054	413
DKT	0.033	0.048	3/5 seeds	-0.063	0.030	413

Note: On seed 42, a KC-clustered percentile bootstrap ($B = 2000$) for SimpleKT ΔFM is $[0.006, 0.138]$; the event-level interval is $[0.015, 0.130]$. Both exclude 0. DKT ΔFM is not stable across seeds and is not treated as a five-run finding. Seeds 2025 and 2026 share a split; both SimpleKT ΔFM values on that split are positive.

4.6 RQ3: Explanatory Conditions

In brief, a sparse-stratum ECE rise is estimable when the split actually contains a low-frequency tail with enough test events and a non-inverted frequency–difficulty coupling (Section 5.4). Between-KC regressions still associate $\log(1 + f_{\text{train}})$ with ECE after training-only covariates (Section 5.3). Holding KC identity fixed and reducing training rows does not reproduce that rise on ASSISTments, but does for SimpleKT on Junyi Academy (Appendix G). Sparse-concept *claims* are high-priority only when those conditions, or cold-start exposure, are present (Section 5.6).

RQ2 shows that a sparse-stratum ECE rise is visible on some datasets and absent on others. RQ3 asks when that rise is estimable at all. Using only training-fold frequencies,

training-only KC covariates, and the event-level SimpleKT ECE already reported in Table 12, Section 5.4 isolates three measurable pre-conditions: (1) a non-empty low-frequency tail under the split and thresholds actually used; (2) enough sparse test events for at least a Limited-flag ECE; and (3) a frequency–difficulty coupling that is not inverted. ASSISTments 2012 meets all three and shows a monotonic ECE increase ($0.114 \rightarrow 0.154 \rightarrow 0.228$). Junyi Academy fails (1) and (2) on the learner-based split, so the protocol leaves sparse ECE blank rather than imputing stability. XES3G5M meets (1) and (2) but not (3); SimpleKT ECE is flat, while IRT is inverted and DKT increases. These are diagnostic pre-conditions, not a claim that training frequency causes miscalibration. Appendix G further tests a controlled reduction of training rows for the same originally dense KCs: a within-KC ECE increase appears in some dataset–model cells (notably SimpleKT on Junyi Academy) and is absent in others (DKT and SimpleKT on ASSISTments 2012). Observable KC characteristics (Section 5.3) explain part of the observational heterogeneity (Finding 4); they do not reduce it to a single frequency dose–response.

That answer is still about *observed vulnerability*. Whether sparse-concept diagnostics are *worth running as a high-priority analysis* additionally depends on evaluation support, intended use of probabilities, and temporal cold-start exposure. Section 5.6 states that distinction as a conditional-use framework (Tables 16 and 17). Occupancy reporting remains useful even when the empirical priority of sparse-stratum claims is low.

4.7 RQ4: Limited Cold-start Concept Diagnostics

How do baseline KT models behave on KCs with zero or limited training-fold frequency under temporal splits? We address this by evaluating models on limited cold-start concepts (strict, k5, k10, warm) under temporal splitting in Table 9.

Table 9: Cold-start Performance Metrics under Temporal Validation

Dataset	Model	Group	#KCs	#Events	AUC	ACC	ECE	Brier	REL	RES
ASSISTments 2012	IRT	strict	27	1,407	0.5000	0.5039	0.1987	0.2895	0.0395	0.0000
		k5	31	1,420	0.4937	0.5049	0.1998	0.2889	0.0403	0.0014
		k10	33	1,713	0.4070	0.4892	0.2143	0.2800	0.0523	0.0172
		warm	212	529,786	0.5919	0.6914	0.0207	0.2080	0.0008	0.0069
	DKT	strict	27	1,407	0.5055	0.5139	0.0598	0.2523	0.0046	0.0026
		k5	31	1,420	0.5093	0.5162	0.0598	0.2517	0.0051	0.0037
		k10	33	1,713	0.4910	0.5096	0.1225	0.2801	0.0529	0.0180
		warm	212	529,786	0.6606	0.6920	0.0844	0.2109	0.0117	0.0150
	SimpleKT	strict	27	1,407	0.4318	0.4634	0.1856	0.3005	0.0604	0.0101
		k5	31	1,420	0.4397	0.4676	0.1845	0.2988	0.0586	0.0099
		k10	33	1,713	0.4153	0.4530	0.2526	0.3456	0.1224	0.0220
		warm	212	529,786	0.6734	0.6804	0.1470	0.2273	0.0304	0.0172
Junyi Academy	IRT	strict	4	2,545	0.5000	0.5147	0.1934	0.2872	0.0374	0.0000
		k5	4	2,545	0.5000	0.5147	0.1934	0.2872	0.0374	0.0000
		k10	4	2,545	0.5000	0.5147	0.1934	0.2872	0.0374	0.0000
		warm	1,319	3,240,570	0.6526	0.7159	0.0220	0.1946	0.0007	0.0162
	DKT	strict	4	2,545	0.5903	0.5611	0.0494	0.2465	0.0051	0.0080
		k5	4	2,545	0.5903	0.5611	0.0494	0.2465	0.0051	0.0080
		k10	4	2,545	0.5903	0.5611	0.0494	0.2465	0.0051	0.0080
		warm	1,319	3,240,570	0.6949	0.7150	0.0746	0.1982	0.0092	0.0213
	SimpleKT	strict	4	2,545	0.5053	0.5501	0.0841	0.2536	0.0155	0.0106
		k5	4	2,545	0.5053	0.5501	0.0841	0.2536	0.0155	0.0106
		k10	4	2,545	0.5053	0.5501	0.0841	0.2536	0.0155	0.0106
		warm	1,319	3,240,570	0.7167	0.7199	0.0911	0.1945	0.0109	0.0264
XES3G5M	IRT	strict	117	233,214	0.5000	0.7875	0.2193	0.2154	0.0481	0.0000
		k5	125	258,943	0.5156	0.7691	0.2295	0.2177	0.0539	0.0009
		k10	126	259,425	0.5162	0.7689	0.2294	0.2177	0.0538	0.0009
		warm	424	1,331,318	0.6561	0.7822	0.0670	0.1590	0.0091	0.0074
	DKT	strict	117	233,214	0.4642	0.5010	0.2812	0.2549	0.0878	0.0007
		k5	125	258,943	0.4778	0.5253	0.2686	0.2472	0.0832	0.0011
		k10	126	259,425	0.4780	0.5257	0.2684	0.2471	0.0831	0.0011
		warm	424	1,331,318	0.6626	0.7456	0.1437	0.1875	0.0394	0.0092
	SimpleKT	strict	117	233,214	0.4773	0.5011	0.2973	0.2896	0.1223	0.0004
		k5	125	258,943	0.5015	0.5329	0.2774	0.2747	0.1109	0.0011
		k10	126	259,425	0.5017	0.5332	0.2773	0.2747	0.1108	0.0011
		warm	424	1,331,318	0.6615	0.7415	0.1860	0.2098	0.0607	0.0079

Note: Cold-start groups are defined using training-fold KC frequency: strict = 0, $k5 \leq 5$, $k10 \leq 10$, and warm is the complementary group. For Junyi Academy, strict, k5, and k10 coincide because the same four concepts have zero training frequency. Warm-cohort temporal AUC serves as the L8 predictive-sanity reference; strict and limited-frequency groups should be interpreted together with sample size and alignment-audit evidence.

Temporal splits expose a demanding generalization setting. Both DKT and SimpleKT generalize poorly to KCs with zero or limited training-fold history, yielding highly degraded, mostly near-random or worse-than-random performance ($AUC \approx 0.41$ – 0.59) on strict and limited cold-start groups (e.g., DKT 0.5055 and SimpleKT 0.4318 on ASSISTments 2012 strict cold-start)—expected, since the models lack parameter history for unseen concepts. Notably, on XES3G5M, the strict cold-start stratum encompasses a massive 233,214 test events across 117 KCs (carrying the Reliable flag R). Despite this large sample size, both deep models systematically yield worse-than-random discrimination (DKT: 0.4642; SimpleKT: 0.4773). With such a large N , this underperformance is unlikely to be attributable to statistical noise. One plausible explanation—consistent with, but not directly evidenced by, our current diagnostics—is a difficulty confound in which unseen concepts in this dataset follow a different difficulty distribution from the training concepts. Confirming this would require item-level difficulty analysis, which lies beyond the scope of this diagnostic protocol and is left to future work. For IRT, the model lacks difficulty parameters for unseen con-

cepts; our implementation thus falls back to a constant base-rate prediction, yielding exactly $AUC = 0.5000$ on the strict cold-start stratum across all datasets, even though the temporal learner abilities are already known from the training phase. In contrast, the limited cold-start groups $k5$ and $k10$ ($f_{\text{train}} \leq 5$ and ≤ 10) additionally include concepts with small but nonzero training frequency, providing just enough interactions for IRT to estimate noisy difficulty parameters for those concepts, resulting in non-constant predictions and AUCs that deviate from 0.5000. After the label-alignment correction of Appendix I was applied consistently across datasets, the deep baselines recover meaningful signal on warm temporal cohorts: DKT and SimpleKT reach 0.6949 and 0.7167 on Junyi Academy, 0.6626 and 0.6615 on XES3G5M, and 0.6606 and 0.6734 on ASSISTments 2012. The contrast is the diagnostic payoff: the dominant temporal difficulty lies in generalizing to unseen or low-frequency KCs, not in forecasting familiar ones, and the warm-cohort recovery is what distinguishes genuine concept-level cold-start failure from pipeline-induced random predictions (channel L8).

L8 is the only audit channel in this study with a positive empirical demonstration rather than a preventative PASS. Before correction, deep KT models produced near-random ranking on *warm* temporal cohorts—KCs with ample training-fold evidence—while IRT on the same cohorts remained informative. That pattern is a pipeline warning, not a scientific result about concept-level cold-start. After pairing predictions and labels by a stable test-instance identity (Appendix I), warm-cohort AUC recovered to the values above. We report this as evidence that an operational predictive-sanity check can prevent a corrupted temporal conclusion; we do not present L1–L8 as a new auditing methodology.

To move beyond a single catch, Appendix J injects six controlled alignment-fault classes into copies of the existing temporal prediction CSVs and applies the L8 rule without retuning thresholds on those faults. Detection was 23/60 injected faults (38.3%), with 0/6 false positives on the clean copies. L8 flagged every label-identity shift (F1, F2) and every global row-index mismatch (F5), and none of the adjacent-timestep prediction shifts (F3). That last miss is expected: learner sequences are autocorrelated, so pairing p_t with a neighboring outcome often leaves warm AUC well above 0.55. L8 is therefore an empirically evaluated sanity mechanism for the fault classes we tested, not a general bug detector.

4.8 Threshold Sensitivity Analysis

Threshold sensitivity analysis is provided as an optional robustness check in the released artifact. The main conclusions rest on the pre-registered KC-frequency thresholds reported in Tables 5, 9, 12, and 13.

5 Discussion and Limitations

In this section, we discuss the broader implications of our diagnostic protocol for educational technology research and practice. We then define the scope boundaries of our work and address the primary limitations of the current study.

5.1 Implications

The protocol encourages KT evaluation to move beyond aggregate AUC in three concrete ways:

- **For researchers:** Per-stratum reporting with reliability flags (Section 3.5) makes claims about sparse and cold-start KCs auditable. A reported gain on low-frequency concepts

is interpretable only together with its stratum sample size, and the protocol makes that context mandatory rather than optional. Crucially, our results do not show a broadly different model choice across strata; for instance, DKT often remains the most discriminative model across frequency buckets. However, we did observe diagnostic disagreements where the best model by AUC is not the best by Expected Calibration Error in every stratum (e.g., SimpleKT outperforming DKT in calibration on the ASSISTments medium stratum). We emphasize that the value of the protocol lies in this granular diagnosis of predictive and probabilistic reliability, not in overturning aggregate model leaderboards.

- **For developers of educational systems:** The calibration diagnostics speak directly to deployment practice. When predicted probabilities drive remediation, practice, or advancement decisions through a *global* threshold, stratum ECE is not the quantity the system consumes—the gate is. On ASSISTments 2012, learner-based fold 0, SimpleKT at $\tau = 0.7$ yields false mastery among advances of 0.196 on dense KCs versus 0.268 on sparse KCs (Limited, $N = 444$; Table 7). Across the five learner-based training seeds, SimpleKT ΔFM remains positive (mean 0.047, 5/5 seeds, four unique partitions; Table 8). The same fold’s train-only GKT instantiation shrinks the seed-42 gap to 0.015. Threshold-based intervention logic should therefore be validated per stratum, not on population-level AUC or ECE alone. The simulation is not a classroom policy.
- **For the reviewing and benchmarking community:** We do not claim that L1–L7 are a new auditing methodology; they are an operational checklist of standard safeguards with named KT artifacts (Table 1). The empirical increment is L8. That channel caught a prediction–label misalignment that aggregate metrics would have misreported as a scientific finding about temporal cold-start difficulty (Section 5.8; Appendix I). A controlled fault-injection validation (Appendix J) subsequently measured sensitivity on six injected fault classes: L8 detected identity-shifted labels and global row-index mismatch, but not adjacent-timestep prediction shifts. We do not claim it detects every bug. An operational checklist with predictive-sanity validation is inexpensive relative to the cost of a corrupted conclusion entering the literature.

5.2 Calibration Can Be More Sensitive Than Discrimination in Some Sparse-Concept Regimes

Our results indicate that the most pronounced sparse-concept vulnerability is not a universal loss of discriminative power (AUC). On ASSISTments 2012, SimpleKT ECE rises from 0.1136 on dense KCs to 0.2280 on sparse KCs (Limited flag). On other datasets the stratum-level ECE pattern is flat or inverted. Calibration can therefore be a more sensitive diagnostic dimension than aggregate discrimination *in some sparse-concept regimes*, but the calibration–frequency relationship is dataset-dependent. One cannot assume that calibration transfers safely to sparse concepts, nor that sparse-stratum ECE must rise. Section 4.5 shows the deployment reading of that gradient when a global τ is applied. On the seed-42 ASSISTments fold, the sparse ECE itself is also lower for a train-only GKT graph (0.098) than for SimpleKT (0.210).

5.3 What Explains Calibration Degradation Across KC-Frequency Strata?

A natural concern is that the calibration pattern observed across KC-frequency strata may reflect differences in difficulty, curriculum position, item structure, or concept tagging rather than training frequency itself. To address this directly, we conducted a KC-level confounding analysis using exclusively training-split information to construct all explanatory variables, thereby preserving the no-leakage guarantee of the protocol.

Covariates

For each KC, we extracted the following training-only characteristics: (i) training frequency, $\log(1 + f_{\text{train}})$; (ii) a difficulty proxy, $1 - \bar{c}_{\text{train}}$ (the complement of mean correctness on the training fold; this is an observational proxy and is *not* a latent IRT difficulty estimate); (iii) the number of distinct items associated with the KC in training ($N_{\text{items/KC}}$); (iv) the number of distinct learners exposed in training (N_{learners}); and (v) a curriculum-position proxy (median normalized sequence position across training interactions).

Univariate Associations

Spearman correlations between $\log(1 + f_{\text{train}})$ and ECE at the KC level were negative in all three datasets and all three models (all $p < 0.001$): ASSISTments 2012 ($\bar{\rho} = -0.60$, strong); Junyi Academy ($\bar{\rho} = -0.43$, moderate); XES3G5M ($\bar{\rho} = -0.23$, weak). The negative sign indicates that lower training frequency is associated with higher ECE. Difficulty (as proxied by $1 - \bar{c}_{\text{train}}$) also showed a positive and significant correlation with ECE in each dataset ($\bar{\rho} \in [0.19, 0.48]$), while curriculum position showed weaker and less uniform associations.

Multivariable Adjustment

We fit weighted linear regressions (weights proportional to test-event count) with ECE as outcome and all five covariates entered simultaneously, estimated separately per dataset with SimpleKT as the primary model of interest. A sensitivity analysis with unweighted OLS yielded coefficients of the same sign. After adjusting for difficulty, curriculum position, item structure, and number of learners, $\log(1 + f_{\text{train}})$ remained independently associated with ECE in all three datasets (ASSISTments 2012: $\hat{\beta} = -0.079$, $p < 0.001$; Junyi Academy: $\hat{\beta} = -0.010$, $p < 0.001$; XES3G5M: $\hat{\beta} = -0.117$, $p < 0.001$, weighted). Difficulty was also independently associated ($p < 0.001$ in ASSISTments 2012 and Junyi Academy), confirming that both frequency and difficulty are contributing factors. In XES3G5M, the unweighted analysis attenuated the frequency coefficient ($p = 0.18$), suggesting that the frequency association in that dataset is partly driven by high-event KCs, consistent with the heterogeneous stratum patterns we report in Table 5.

Matched Analysis Within Difficulty Strata

In the difficulty-stratified matched analysis, lower-frequency KCs had significantly higher ECE than higher-frequency KCs within the same difficulty tertile in 8 of 9 stratum comparisons across datasets (ASSISTments 2012: all 3 tertiles significant, $\Delta\text{ECE} \in [0.021, 0.131]$; Junyi Academy: all 3 tertiles significant, $\Delta\text{ECE} \in [0.003, 0.022]$; XES3G5M: 2 of 3 tertiles significant). This matched analysis controls for difficulty at the KC level and provides evidence that the frequency-ECE association is not merely an artefact of easier KCs having higher training frequency.

Summary

Taken together, these analyses indicate that the calibration gradient observed across KC-frequency strata is not fully explained by the observable KC characteristics we can construct from the training fold. Training frequency ($\log f_{\text{train}}$) retains an independent *between-KC* association with ECE after adjustment in all three datasets, though the magnitude varies (strongest in ASSISTments 2012 and XES3G5M, weaker in Junyi Academy). Difficulty is an additional associate. That associational estimand compares different KCs. It does not test what happens when training count is reduced for the *same* KC (Section 5.5). Both questions are needed: the first speaks to observational sparse-stratum gradients, the second to a frequency dose on a fixed KC identity. Practitioners should apply the diagnostic protocol to their own datasets rather than assuming either pattern will generalise from any single dataset. The next subsection replaces the residual “dataset-dependent” clause with measurable structural conditions.

5.4 Why Sparse-Concept Calibration Vulnerability Is Dataset-Dependent

Section 5.3 showed that $\log(1 + f_{\text{train}})$ remains independently associated with ECE after covariate adjustment, but that the strength of this association—and the visibility of a stratum-level ECE gradient—varies across datasets. The remaining question is not whether “datasets differ,” but whether those differences correspond to measurable structural conditions. Table 15 reports six such conditions, all computed from training-fold frequencies, training-only KC covariates, and the event-level SimpleKT ECE already reported in Table 12. Each cell is a FINDING (a structural or metric observation). Explanations that would require KC ontologies, curriculum graphs, or item-text features—hierarchical placement, ceiling effects, item-feature richness, and tagging granularity—are labelled HYPOTHESIS and are not treated as established.

ASSISTments 2012: why a calibration gradient is observable

Under the learner-based split, 18.9% of ASSISTments KCs fall in sparse, very-sparse, or strict buckets, and the sparse stratum accumulates $N = 415$ test events (Limited flag). Event-level SimpleKT ECE increases monotonically from 0.114 (dense) to 0.154 (medium) to 0.228 (sparse). Two training-only couplings have the direction that would be expected if rarer concepts were also harder and later: frequency–difficulty $\rho = -0.227$ ($p < 0.001$) and frequency–curriculum-position $\rho = -0.308$ ($p < 0.001$). The frequency–ECE association is the strongest of the three datasets ($\bar{\rho} = -0.60$). These are jointly the conditions under which a sparse-KC calibration gradient is *estimable and visible* in this dataset.

They do not, however, identify a single cause. Item-support imbalance (median 205 items/KC in dense versus 1 in sparse) is partly mechanical: a KC with $f_{\text{train}} < 100$ cannot host hundreds of training items. The non-mechanical item fact is that ASSISTments KCs are item-rich overall (median 44.5 items/KC; frequency–item $\rho = 0.92$). **HYPOTHESIS:** the relatively small KC inventory ($n = 265$) and right-skewed frequencies may reflect a coarse expert-tagged skill ontology in which low-frequency labels are genuinely rare or advanced skills. Confirming that claim would require external KC metadata, which we do not use.

Junyi Academy: why learner-based sparse strata are inactive

Junyi has 1,326 KCs, but 97.5% are dense and 0.0% are sparse or very sparse under the pre-registered thresholds $\{20, 100, 500\}$. Median training frequency is 7,074—about seven

times the ASSISTments median (988). Consequently the learner-based sparse and very-sparse ECE cells are empty, not because calibration is known to be stable, but because the protocol’s buckets receive no mass. This is a FINDING about volume and thresholds, not about the absence of hard-to-calibrate concepts. The same protocol populates sparse-like groups under the temporal split (10 KCs with $f_{\text{train}} < 100$, and a SimpleKT sparse ECE of 0.1624 versus dense 0.0889 in Table 13).

Where Junyi *does* have contrast (dense versus medium), the direction matches ASSISTments: medium ECE is higher ($\Delta\text{ECE} = +0.028$), frequency–difficulty coupling is present and in the expected direction ($\rho = -0.416$), and the frequency–ECE association is moderate ($\bar{\rho} = -0.43$). Junyi item support is unusually uniform (median 18 items/KC, IQR 5; frequency–item $\rho = 0.026$, $p = 0.34$), so item-count heterogeneity is not a plausible driver of the medium-stratum ECE rise. **HYPOTHESIS:** raising the dense threshold toward the Junyi frequency scale (for example $f_{\text{train}} \geq 2000$) might reopen a sparse-like contrast under learner-based splits. That is a threshold-sensitivity question, not a result of this study.

XES3G5M: why the pattern is not a second ASSISTments

XES3G5M rules out the two simplest explanations for a missing gradient. It has *more* sparse-like mass than ASSISTments (22.5% of KCs) and *better* sparse test support (sparse $N = 2,010$, Reliable; very-sparse $N = 112$, Limited). A flat or inverted pattern here is therefore not an artefact of empty buckets or of $N < 100$ instability.

What *is* measurable and different: (i) SimpleKT event-level ECE is essentially flat from dense to sparse ($0.114 \rightarrow 0.111 \rightarrow 0.125$), while IRT is inverted ($0.155 \rightarrow 0.112$ dense-to-sparse) and DKT increases—so “the XES3G5M gradient” is model-heterogeneous, not a single dataset law; (ii) the frequency–ECE association is the weakest of the three datasets ($\bar{\rho} = -0.23$); (iii) frequency–difficulty coupling is weak and opposite-signed ($\rho = +0.087$, $p = 0.012$), not a strong “sparse KCs are easier” effect; (iv) KCs are easier overall (difficulty-proxy median 0.192 versus 0.338 on both other datasets); (v) item support is thin (median 3 items/KC) and interaction mass is concentrated (top 10% of KCs hold 67.9% of training interactions; the single most frequent KC holds 19.4%).

HYPOTHESIS: a hierarchical KC tree, multi-skill expansion, a ceiling on advanced nodes, or item-side features could produce this combination. None of those mechanisms is identified by the covariates we can construct from the processed interaction schema, so we do not promote them to findings.

Conditional conclusion

Across the three datasets, an *observable* SimpleKT sparse-KC calibration gradient coincides with three measurable conditions: (1) non-trivial sparse mass under the split and thresholds actually used; (2) enough sparse test events to support at least a Limited-flag ECE; and (3) a frequency–difficulty coupling that is not inverted. ASSISTments 2012 meets all three and shows a monotonic ECE rise. Junyi Academy fails (1) and (2) on the learner-based split, so the protocol correctly withholds a sparse ECE rather than imputing stability. XES3G5M meets (1) and (2) but not (3), and the SimpleKT gradient is flat; model-specific counter-patterns (inverted IRT, increasing DKT) further forbid a universal sparsity-to-miscalibration law. These are diagnostic pre-conditions that future studies can check with the same protocol. They are not a claim that training frequency causes miscalibration, nor that hierarchical curricula, ceiling effects, or tagging granularity have been demonstrated.

5.5 Controlled Reduction of Training Evidence

The between-KC regressions in Section 5.3 still leave open a composition concern: sparse KCs may differ in identity, tagging, and item structure, not only in f_{train} . Those two analyses answer different questions. Section 5.3 asks whether KCs that *already* have lower training counts also have higher ECE, after other training-only covariates. Appendix G asks whether reducing training rows for the *same* originally dense KC is followed by a within-KC ECE increase. A positive association in the first analysis and a null (or opposite) within-KC change in the second are compatible: the observational gradient can be driven partly by which KCs are sparse, while a frequency dose on a fixed KC can still appear in other dataset–model cells.

Appendix G therefore downsamples training rows for 30 originally dense KCs per dataset—selected by a pre-registered rule that does not use ECE—while holding the test set, labels, KC identifiers, and all other KCs’ training rows fixed. DKT and SimpleKT were retrained on ASSISTments 2012, Junyi Academy, and XES3G5M (learner-based fold 0, seed 42). The 100% control is the published prediction export, not a second full-fold retrain; reduced runs use validation-based early stopping.

The primary within-KC result is dataset- and model-dependent (Table 18). On ASSISTments 2012, mean ΔECE was negative for DKT at 500 and 100 kept rows (bootstrap CIs exclude 0) and indistinguishable from 0 for SimpleKT at 100 and 50 rows. On Junyi Academy, DKT showed the same pattern, but SimpleKT ΔECE was large and positive at every level (0.101 to 0.135; CIs exclude 0; 93–100% of selected KCs worse). On XES3G5M, SimpleKT ΔECE was smaller but positive at 100 and 50 rows; DKT increased only at 50 rows. Selected-KC discrimination fell in every cell (e.g. ASSISTments DKT AUC 0.690 $\rightarrow$ 0.627; SimpleKT 0.679 $\rightarrow$ 0.607 from full to 50 rows). Event-pooled ECE, as in Figure 4, can move differently from the within-KC mean when a few high-volume KCs dominate (on ASSISTments SimpleKT the pooled ECE rises while mean within-KC ΔECE does not); the pre-registered primary outcome remains the paired within-KC $\Delta\text{ECE}/\Delta\text{REL}$.

We do not conclude that real-world sparsity always causes miscalibration. Reducing training rows for the same originally dense KCs can coincide with a within-KC ECE increase, but only in some dataset–model cells. The ASSISTments observational dense-to-sparse SimpleKT gradient (0.114 $\rightarrow$ 0.228 in Table 12) was not recreated by sparsifying those ASSISTments KCs, so that particular observational pattern is still likely explained partly by KC composition rather than by training count alone. The Junyi SimpleKT within-KC increase shows that a frequency reduction on a fixed KC is nevertheless associated with worse ECE in at least one instantiation.

5.6 When Are Sparse-Concept Diagnostics Necessary?

RQ3 answers when a sparse-stratum calibration gradient is *estimable*. That is not the same question as whether every KT dataset needs the same intensity of sparse-concept analysis. We separate two layers.

Reporting hygiene. Train-only frequency occupancy, empty-bucket notes, and reliability flags (R/L/I) should be reported whenever KC-frequency strata are defined. An empty sparse stratum is an informative outcome: it tells the reader that sparse-ECE claims are not supported under the chosen split and thresholds. Junyi Academy under the learner-based split is the example in this paper. Hygiene of this kind does not require a calibration gradient, and it does not license the conclusion that probabilities are well calibrated on rare concepts.

Empirically high-priority sparse diagnostics. Claims about sparse-KC ECE, Brier reliability, or cold-start AUC become high-priority when the data can actually support them. Table 16 states that condition as five checks (C1–C5). C1–C3 and C5 reuse the protocol’s pre-registered sparse threshold ($f_{\text{train}} < 100$) and reliability flags; they are not new cutoffs fitted to these three datasets. C4 is qualitative intended use: it is High when predicted probabilities gate remediation, practice, or mastery, and Low when only rankings are consumed. We do not estimate C4 from the public logs. Section 4.5 instead *assumes* the High setting and simulates a global τ on existing test probabilities; that exercise measures decision error under the assumption, not whether a deployed system actually thresholds p .

Table 17 applies the framework. ASSISTments 2012 is high priority for sparse-concept calibration claims: 18.9% of KCs fall below the sparse threshold, the sparse stratum has Limited support ($N = 415$), and SimpleKT ECE rises monotonically. Junyi Academy is low priority for learner-based sparse-stratum ECE (the buckets are empty) and only moderate under the temporal split, where 10 sparse-like KCs appear and sparse ECE is 0.1624 versus dense 0.0889. XES3G5M shows why need is not the same as vulnerability: sparse mass (22.5%) and Reliable sparse support ($N = 2,010$), plus 233,214 strict cold-start test events, make the diagnostics necessary as reporting; the same diagnostics show a flat SimpleKT ECE and an inverted IRT pattern, not an ASSISTments-like degradation. If probabilities were used as educational thresholds (C4), that reporting priority would rise on all three datasets alike. The ASSISTments simulation in Section 4.5 is what that High-C4 setting looks like as a number; Junyi cannot score a sparse gate under the learner-based split, and XES3G5M shows that a flat SimpleKT ECE can coexist with a larger sparse–dense miss gap.

The conditional statement is therefore: sparse diagnostics are most informative when a dataset contains a meaningful low-frequency tail, those strata have sufficient evaluation support, and calibration reliability varies with training evidence or deployment involves concept cold-start. They are not equally informative on every KT log, and occupancy hygiene remains useful even when the empirical priority is low.

5.7 Scope Boundaries

This paper is intentionally limited to evaluation diagnostics. It does not introduce a new KT model, a self-supervised learning (SSL) objective, a graph neural network (GNN), a graph augmentation strategy, a learning path recommender (LPR), or a distillation method. The GKT run and CL4KT adapter in Section 4.5 are protocol instantiations of published architectures, not proposed methods, and they are reported on a single ASSISTments fold. Those directions still require separate methodological papers if the goal is a new learner model. A complementary line of work addresses leakage control in concept-graph construction for graph-augmented KT; the train-only versus full-log GKT ablation is one concrete illustration of that complementarity.

5.8 Threats to Validity and Limitations

We analyze the threats to the validity of our diagnostic protocol across standard dimensions:

- **Internal Validity:** Risks to internal validity include potential preprocessing bugs, split leakage, sequence–label alignment errors, and implementation-specific training differences across the baselines. We mitigate these risks through a standardized interaction data schema, rigid validation-fold early stopping, and the L1–L8 reproducible audit workflow. Additionally, we note that our classical baseline (IRT) under learner-based

splits acts as a base-rate reference due to the lack of ability estimates for unseen learners; developing a classical baseline that effectively exploits item difficulty for unseen learners would strengthen future comparative work.

- **Temporal Alignment Validity:** Sequence modeling in KT is notoriously prone to “silent bugs” such as off-by-one prediction–label misalignments. Our protocol actively defends against this via the warm-cohort predictive sanity audit (L8). During validation, this automated check flagged an alignment discrepancy in the temporal evaluation pipeline that otherwise would have corrupted the evaluation. The subsequent fault-injection study (Appendix J) shows that L8 is sensitive to identity-shifted labels and row-index mismatch, but not to adjacent-timestep prediction shifts that remain autocorrelated; L8 is not claimed to catch every alignment bug.
- **Construct Validity:** Construct validity refers to whether our metrics accurately capture latent learner mastery and concept calibration. In KT, KC definitions depend heavily on dataset annotation quality, Q-matrix provenance, and how multi-skill items are processed. For example, expert-tagged KCs in ASSISTments 2012 and Junyi Academy may introduce varying granularities, which directly impact strata boundaries.
- **External Validity:** Section 5.4 shows that the three datasets differ for measurable structural reasons (sparse mass, test-event support, and frequency–difficulty coupling), not merely as an unexplained residual. Section 5.6 turns those conditions into a use framework: sparse-concept diagnostic *claims* are high-priority only when the tail, the evaluation support, and either a calibration association or cold-start exposure are present. Occupancy reporting remains useful as reporting hygiene on every dataset we consider. Those conditions are themselves dataset- and pipeline-specific, so models still cannot be assumed to share a sparsity-to-miscalibration law.
- **Statistical Conclusion Validity:** very sparse buckets can contain few KCs and few evaluation events (as few as 3 to 10 events per seed in our instantiation), which inevitably produces metric instability and high variance; the reliability flags of Section 3.5 exist precisely to keep such estimates from being over-interpreted. Two further limitations belong under this heading. First, the temporal-split diagnostics are computed from a single corrected evaluation run at a fixed chronological cutoff (Section 4.1); they therefore characterize a fixed-cutoff stress test and do not estimate variance across temporal cut-points or neural initializations. Extending the temporal diagnostics to multiple seeds and multiple cutoffs is a direct avenue for future work. Second, because the Brier decomposition is computed from binned probabilities (Section 3.4), its components are empirical approximations and need not sum exactly to the directly computed Brier score; decomposition values should be read as diagnostic indicators rather than exact accounting identities. The controlled sparsification experiment (Section 5.5) is likewise a single-seed (42) comparison against the published full-fold export; reduced-evidence runs use early stopping, so ΔECE is not a matched-epoch retrain of the control. The GKT, CL4KT-adaptor, and seed-42 threshold-simulation details (Section 4.5) use a single fold; sparse FM is a Limited-flag proportion. SimpleKT ΔFM at $\tau = 0.7$ is additionally reported across five training seeds (four unique partitions; Table 8). Learner-based AUC/ECE tables summarize four unique partitions (the duplicated 2025/2026 split is averaged first). The CL4KT numbers must not be read as a replication of an official CL4KT release.

6 Conclusion

This paper presented a diagnostic protocol for sparse-concept and calibration evaluation in Knowledge Tracing: train-only KC-frequency stratification with an explicit strict cold-start group, learner-based and temporal splits, per-stratum calibration analysis (ECE and the Brier decomposition, with reliability diagrams), sample-size-aware reliability flags, and a reproducible audit workflow (L1–L8) that operationalizes standard safeguards plus a predictive-sanity check. The protocol is intended as reusable evaluation infrastructure for KT research, including graph-based and self-supervised models whose claimed gains concentrate on sparse KCs, where evaluation is least stable. Section 4.5 is a single-fold illustration of that use, not a new architecture paper.

The instantiation on three datasets supports a bounded scientific story. Sparse training evidence does not universally degrade discrimination, but it can expose dataset-dependent calibration vulnerability. We characterize the conditions associated with this vulnerability—sparse mass, evaluation support, and frequency–difficulty coupling—and provide reproducible diagnostics for identifying when it matters. On ASSISTments 2012, SimpleKT ECE rises from 0.114 on dense concepts to 0.154 on medium concepts and 0.228 on sparse concepts (Limited flag, $N = 415$). Other datasets show flat or inverted stratum patterns. On the seed-42 learner-based fold of the same dataset, a global gate at $\tau = 0.7$ raises SimpleKT false mastery among advances from 0.196 (dense) to 0.268 (sparse); that ΔFM is positive on all five training seeds (mean 0.047; four unique partitions). A train-only GKT graph shrinks the seed-42 gap, and a CL4KT adapter is intermediate. This is a simulated decision, not a classroom intervention. A controlled downsampling of training rows for the same originally dense KCs produces a within-KC ECE increase only in some dataset–model cells (Section 5.5), so observational frequency–calibration gradients should not be read as a universal frequency dose–response. Sparse-concept diagnostics are most informative when a dataset contains a meaningful low-frequency tail, those strata have sufficient evaluation support, and calibration reliability varies with training evidence or deployment involves concept cold-start (Section 5.6). Occupancy reporting remains useful even when that empirical priority is low. Claim intensity is therefore conditional: not every KT log should be narrated as a sparse-calibration failure. Finally, channel L8 showed empirical value: it caught a prediction–label alignment artifact that would otherwise have been misread as genuine cold-start failure, and a controlled fault-injection validation later measured sensitivity to additional alignment-fault classes (Appendix J). L8 is reported as sensitive to the classes we tested, not as a detector of every evaluation bug. L1–L7 remain standard safeguards made inspectable; they are not offered as a new auditing methodology.

Future work can extend this diagnostic foundation by evaluating across multiple temporal seeds or cutoffs to capture variance, by incorporating a classical baseline that exploits item difficulty for unseen learners under learner-based splits, by applying post-hoc or stratum-aware calibration (a separate methods question), and by repeating the GKT/CL4KT occupancy-and-gate checks on additional datasets and official checkpoints.

Artifact Availability & Reproducibility Checklist

The anonymized code, data preprocessing scripts, and reproducible evaluation pipelines are available for peer review at <https://anonymous.4open.science/r/Sparse-Concept-and-Calibration-6E5B/>. The public repository will be released upon acceptance. Consistent with the design principles of Section 3, the artifact

includes the following standardized components:

- **Dataset Preprocessing:** Scripts that normalize raw data into the schema defined in Section 3.
- **Split Construction:** Reproducible partition logs and multi-seed split constructors.
- **Prediction-level Exports:** CSV files containing baseline predicted probabilities and ground-truth labels.
- **KC-Strata Assignment:** Code mapping training-fold frequencies to strata buckets.
- **Calibration Verification:** Scripts computing Expected Calibration Error (ECE).
- **Brier Decomposition:** Implementation of Uncertainty, Reliability, and Resolution equations.
- **Reliability Diagrams:** Plotting utilities for multi-strata calibration curves.
- **Leakage & Predictive-Sanity Workflow:** Automated logs for the L1–L8 operational checklist, including the L8 warm-cohort sanity check.
- **LaTeX Table Exporters:** Python exporters generating booktabs tables.
- **One-Command Reproduction:** Unified script ensuring machine-independent reproducibility.

A Threshold Sensitivity Protocol

Threshold sensitivity analysis is implemented in the released artifact as an optional robustness check. The present manuscript does not use threshold-sensitivity results as primary evidence because the main diagnostic conclusions are based on the pre-registered KC-frequency thresholds used in Tables 5, 6, 9, and 10. The artifact evaluates three alternative threshold configurations: (i) an aggressive setting with thresholds $\{10, 50, 250\}$ training interactions, (ii) a conservative setting with thresholds $\{30, 150, 750\}$ training interactions, and (iii) a dynamic quantile-based bucketing scheme where buckets are defined by equal-quantile KCs. These checks are intended to help future users evaluate whether their conclusions are sensitive to the chosen KC-frequency thresholds.

B Overall and Bucket-level Performance under Temporal Splits

In this section, we present the detailed overall performance (Table 10) and strata-specific performance breakdowns (Table 11) under the time-partitioned (temporal) split mode. As shown, deep learning baselines undergo noticeable predictive decay compared to their population-level learner-based counterparts, highlighting the increased difficulty of curriculum-level forecasting. Calibration degradation under temporal splits is reported separately in Appendix C, Table 13.

Table 10: Overall Performance under Future Validation (Temporal Splits)

Dataset	Split	Model	AUC	ACC	NLL	RMSE
ASSISTments 2012	Temporal	IRT	0.5913	0.6908	0.6049	0.4563
	Temporal	DKT	0.6606	0.6915	0.6362	0.4595
	Temporal	SimpleKT	0.6731	0.6796	0.7648	0.4771
Junyi Academy	Temporal	IRT	0.6525	0.7157	0.5738	0.4412
	Temporal	DKT	0.6949	0.7148	0.6181	0.4452
	Temporal	SimpleKT	0.7167	0.7198	0.6320	0.4411
XES3G5M	Temporal	IRT	0.6316	0.7801	0.5137	0.4106
	Temporal	DKT	0.6429	0.7097	0.6429	0.4441
	Temporal	SimpleKT	0.6433	0.7075	0.9014	0.4694

Table 11: Knowledge Tracing Performance Breakdown by KC-frequency Strata (Temporal Splits)

Dataset	Model	Bucket	Rel.	#KCs	#Events	AUC	ACC	NLL	RMSE
ASSISTments 2012	IRT	dense	R	145	521,661	0.5917	0.6929	0.6032	0.4554
		medium	R	29	4,558	0.5332	0.6038	0.6797	0.4924
		sparse	R	28	3,030	0.6486	0.5891	0.6789	0.4919
		Very Sparse	L	16	843	0.7380	0.5302	0.6680	0.4875
		Strict Cold-start	R	27	1,407	0.5000	0.5039	0.7794	0.5380
	DKT	dense	R	145	521,661	0.6619	0.6942	0.6256	0.4570
		medium	R	29	4,558	0.5637	0.5432	1.1556	0.5818
		sparse	R	28	3,030	0.6244	0.5743	1.3584	0.5859
		Very Sparse	L	16	843	0.5971	0.4864	1.6419	0.6299
		Strict Cold-start	R	27	1,407	0.5055	0.5139	0.6979	0.5023
	SimpleKT	dense	R	145	521,661	0.6745	0.6818	0.7556	0.4753
		medium	R	29	4,558	0.6084	0.5838	0.9636	0.5420
		sparse	R	28	3,030	0.6117	0.6023	1.6744	0.5800
		Very Sparse	L	16	843	0.5617	0.5006	2.0213	0.6459
		Strict Cold-start	R	27	1,407	0.4318	0.4634	0.8124	0.5482
Junyi Academy	IRT	dense	R	1,235	3,072,767	0.6483	0.7207	0.5692	0.4389
		medium	R	78	151,597	0.6487	0.6219	0.6582	0.4825
		sparse	R	6	16,206	0.6173	0.6703	0.6291	0.4679
		Strict Cold-start	R	4	2,545	0.5000	0.5147	0.7752	0.5359
	DKT	dense	R	1,235	3,072,767	0.7073	0.7238	0.5784	0.4360
		medium	R	78	151,597	0.5562	0.5489	1.3060	0.5886
		sparse	R	6	16,206	0.5183	0.5930	1.6870	0.5857
		Strict Cold-start	R	4	2,545	0.5903	0.5611	0.6862	0.4965
	SimpleKT	dense	R	1,235	3,072,767	0.7178	0.7259	0.6235	0.4369
		medium	R	78	151,597	0.6436	0.6046	0.7854	0.5122
		sparse	R	6	16,206	0.6529	0.6562	0.7917	0.4972
		Strict Cold-start	R	4	2,545	0.5053	0.5501	0.7007	0.5036
XES3G5M	IRT	dense	R	340	1,191,744	0.6673	0.7993	0.4744	0.3893
		medium	R	72	129,494	0.5608	0.6227	0.6488	0.4778
		sparse	R	11	10,079	0.6903	0.8187	0.4720	0.3860
		Very Sparse	R	10	26,212	0.6774	0.6037	0.6713	0.4879
		Strict Cold-start	R	117	233,214	0.5000	0.7875	0.6236	0.4641
	DKT	dense	R	340	1,191,744	0.6679	0.7580	0.6029	0.4227
		medium	R	72	129,494	0.6214	0.6259	0.9269	0.5215
		sparse	R	11	10,079	0.6490	0.8228	0.5643	0.3785
		Very Sparse	R	10	26,212	0.5529	0.7460	0.5518	0.4217
		Strict Cold-start	R	117	233,214	0.4642	0.5010	0.7032	0.5049
	SimpleKT	dense	R	340	1,191,744	0.6666	0.7517	0.8803	0.4478
		medium	R	72	129,494	0.6136	0.6411	1.3648	0.5472
		sparse	R	11	10,079	0.6911	0.8286	0.9727	0.3862
		Very Sparse	R	10	26,212	0.7308	0.8194	0.5170	0.3772
		Strict Cold-start	R	117	233,214	0.4773	0.5011	0.7925	0.5381

Note: This table reports only KCs with at least one valid evaluated prediction row after temporal test-fold filtering, sequence construction, prediction export, and KC-strata matching. The evaluated temporal KC totals are 245/265 for ASSISTments 2012, 1,323/1,326 for Junyi Academy, and 550/866 for XES3G5M. The remaining KCs have no valid evaluated temporal prediction events after filtering (i.e., no learners interacted with them during the chronological test period) and are therefore omitted from metric rows; their omission does not affect the reported event-level metrics but should be considered when interpreting KC coverage. Within a stratum, evaluated event counts may differ across models. This bidirectional variance occurs because (1) sequence models often drop initial interactions lacking prior state or apply minimum-length filtering, reducing dense-event totals compared to classical IRT; (2) IRT implementations often fail to output predictions for concepts with insufficient training history, while deep models still generate predictions via initialized embeddings, increasing their sparse-event totals; and (3) codebase-specific windowing and padding choices cause further discrepancies.

C Detailed Calibration Results

Detailed calibration results by frequency stratum for the learner-based split are reported in Table 12. The corresponding temporal calibration breakdowns, computed after the label-alignment correction documented in Appendix I and used as the numerical basis for Figure 3, are reported in Table 13.

Table 12: Calibration Breakdown by Frequency Stratum

Dataset	Model	Bucket	Rel.	#Events	ECE	Brier	UNC	REL	RES
ASSISTments 2012	IRT	dense	R	523,971	0.0031 ± 0.0006	0.2103 ± 0.0003	0.2103	0.0000	0.0000
		medium	R	5,963	0.1598 ± 0.0190	0.2742 ± 0.0075	0.2484	0.0258	0.0000
		sparse	L	415	0.0777 ± 0.0555	0.2420 ± 0.0218	0.2337	0.0084	0.0000
		Very Sparse	I	16	0.2244 ± 0.0666	0.2996 ± 0.0261	0.2459	0.0537	0.0000
	DKT	Strict Cold-start	I	4	0.6962	0.4848	0.0000	0.4848	0.0000
		dense	R	523,971	0.0602 ± 0.0022	0.1931 ± 0.0010	0.2103	0.0053	0.0223
		medium	R	5,963	0.1621 ± 0.0256	0.2297 ± 0.0144	0.2484	0.0301	0.0479
		sparse	L	415	0.2333 ± 0.0084	0.2502 ± 0.0098	0.2337	0.0624	0.0451
	SimpleKT	Very Sparse	I	16	0.1782 ± 0.0666	0.1464 ± 0.0786	0.2459	0.0862	0.1849
		Strict Cold-start	I	4	0.5014	0.2540	0.0000	0.2539	0.0000
		dense	R	523,971	0.1136 ± 0.0066	0.2116 ± 0.0019	0.2103	0.0201	0.0186
		medium	R	5,963	0.1541 ± 0.0051	0.2272 ± 0.0086	0.2484	0.0273	0.0478
		sparse	L	415	0.2280 ± 0.0197	0.2525 ± 0.0116	0.2337	0.0594	0.0392
		Very Sparse	I	16	0.2451 ± 0.0511	0.2165 ± 0.0461	0.2459	0.0852	0.1148
		Strict Cold-start	I	4	0.5224	0.2761	0.0000	0.2758	0.0000
Junyi Academy	IRT	dense	R	3,226,541	0.0026 ± 0.0018	0.2078 ± 0.0007	0.2078	0.0000	0.0000
		medium	R	3,706	0.1209 ± 0.0082	0.2579 ± 0.0033	0.2432	0.0147	0.0000
	DKT	dense	R	3,226,541	0.0389 ± 0.0022	0.1804 ± 0.0008	0.2078	0.0022	0.0294
		medium	R	3,706	0.2513 ± 0.0217	0.3077 ± 0.0144	0.2432	0.0741	0.0091
	SimpleKT	dense	R	3,232,614	0.0792 ± 0.0051	0.1884 ± 0.0020	0.2079	0.0084	0.0279
		medium	R	3,836	0.1073 ± 0.0156	0.2307 ± 0.0103	0.2417	0.0149	0.0260
XES3G5M	IRT	dense	R	1,269,345	0.1546 ± 0.0033	0.1861 ± 0.0008	0.1622	0.0239	0.0000
		medium	R	12,889	0.1421 ± 0.0060	0.1897 ± 0.0015	0.1695	0.0202	0.0000
		sparse	R	1,969	0.1119 ± 0.0096	0.1982 ± 0.0026	0.1856	0.0126	0.0000
		Very Sparse	L	114	0.0850 ± 0.0420	0.2059 ± 0.0118	0.1973	0.0085	0.0000
	DKT	Strict Cold-start	I	5	0.2362 ± 0.1144	0.2269 ± 0.0836	0.1613	0.0656	0.0000
		dense	R	1,268,696	0.0388 ± 0.0023	0.1221 ± 0.0021	0.1614	0.0027	0.0415
		medium	R	12,980	0.0906 ± 0.0172	0.1231 ± 0.0099	0.1681	0.0133	0.0571
		sparse	R	2,010	0.1187 ± 0.0154	0.1412 ± 0.0087	0.1864	0.0195	0.0637
	SimpleKT	Very Sparse	L	112	0.1840 ± 0.0372	0.1806 ± 0.0364	0.2048	0.0542	0.0762
		Strict Cold-start	I	4	0.4472 ± 0.0937	0.2655 ± 0.0391	0.1022	0.2217	0.0583
		dense	R	1,268,696	0.1145 ± 0.0011	0.1547 ± 0.0029	0.1614	0.0179	0.0237
		medium	R	12,980	0.1114 ± 0.0076	0.1351 ± 0.0060	0.1681	0.0169	0.0488
		sparse	R	2,010	0.1248 ± 0.0085	0.1477 ± 0.0050	0.1864	0.0215	0.0593
		Very Sparse	L	112	0.1832 ± 0.0111	0.1727 ± 0.0110	0.2048	0.0538	0.0850
		Strict Cold-start	I	4	0.5738 ± 0.2468	0.4192 ± 0.2186	0.1022	0.3844	0.0661

Note: Reliability flags (R/L/I) follow Section 3.5. Mean $\pm$ standard deviation and event counts are over four unique learner-based partitions (seeds 2025 and 2026 averaged first). IRT shows low ECE in several learner-based cohorts, but RES = 0 across strata indicates base-rate-like behavior with no resolving power (Section 4.2); IRT calibration should therefore be interpreted jointly with AUC and Brier resolution.

Table 13: Calibration Breakdown by Frequency Stratum under Temporal Splits. Values are computed from prediction-level outputs after the label-alignment correction documented in Appendix F. Very Sparse excludes Strict Cold-start KCs; Strict Cold-start is reported separately where applicable. This table provides the source calibration values used for Figure 3.

Dataset	Model	Bucket	Rel.	#Events	ECE	Brier	UNC	REL	RES
ASSISTments 2012	IRT	dense	R	521,661	0.0194	0.2074	0.2138	0.0007	0.0069
		medium	R	4,558	0.0853	0.2425	0.2347	0.0101	0.0027
		sparse	R	3,030	0.1374	0.2419	0.2340	0.0261	0.0179
		Very Sparse	L	843	0.2492	0.2376	0.2042	0.0674	0.0331
		Strict Cold-start	R	1,407	0.1987	0.2895	0.2500	0.0395	0.0000
	DKT	dense	R	521,661	0.0812	0.2088	0.2138	0.0104	0.0154
		medium	R	4,558	0.2796	0.3385	0.2347	0.1068	0.0036
		sparse	R	3,030	0.3149	0.3433	0.2340	0.1228	0.0127
		Very Sparse	L	843	0.3553	0.3967	0.2042	0.2100	0.0162
		Strict Cold-start	R	1,407	0.0598	0.2523	0.2500	0.0046	0.0026
	SimpleKT	dense	R	521,661	0.1454	0.2260	0.2138	0.0297	0.0174
		medium	R	4,558	0.2277	0.2938	0.2347	0.0677	0.0089
		sparse	R	3,030	0.3069	0.3363	0.2340	0.1120	0.0093
		Very Sparse	L	843	0.3879	0.4172	0.2042	0.2322	0.0189
		Strict Cold-start	R	1,407	0.1856	0.3005	0.2500	0.0604	0.0101
Junyi Academy	IRT	dense	R	3,072,767	0.0233	0.1926	0.2070	0.0008	0.0150
		medium	R	151,597	0.0625	0.2328	0.2479	0.0051	0.0201
		sparse	R	16,206	0.0398	0.2189	0.2306	0.0020	0.0136
		Strict Cold-start	R	2,545	0.1934	0.2872	0.2498	0.0374	0.0000
	DKT	dense	R	3,072,767	0.0629	0.1901	0.2070	0.0062	0.0230
		medium	R	151,597	0.2921	0.3465	0.2479	0.1007	0.0023
		sparse	R	16,206	0.3131	0.3430	0.2306	0.1130	0.0010
		Strict Cold-start	R	2,545	0.0494	0.2465	0.2498	0.0051	0.0080
	SimpleKT	dense	R	3,072,767	0.0889	0.1909	0.2070	0.0102	0.0262
		medium	R	151,597	0.1552	0.2623	0.2479	0.0299	0.0154
		sparse	R	16,206	0.1624	0.2472	0.2306	0.0324	0.0158
		Strict Cold-start	R	2,545	0.0841	0.2536	0.2498	0.0155	0.0106
XES3G5M	IRT	dense	R	1,191,744	0.0547	0.1516	0.1563	0.0046	0.0092
		medium	R	129,494	0.1759	0.2283	0.1691	0.0616	0.0026
		sparse	R	10,079	0.1823	0.1490	0.1202	0.0368	0.0080
		Very Sparse	R	26,212	0.3189	0.2380	0.1404	0.1061	0.0085
		Strict Cold-start	R	233,214	0.2193	0.2154	0.1673	0.0481	0.0000
	DKT	dense	R	1,191,744	0.1315	0.1787	0.1563	0.0324	0.0099
		medium	R	129,494	0.2573	0.2720	0.1691	0.1097	0.0065
		sparse	R	10,079	0.1234	0.1433	0.1202	0.0307	0.0076
		Very Sparse	R	26,212	0.1549	0.1779	0.1404	0.0420	0.0047
		Strict Cold-start	R	233,214	0.2812	0.2549	0.1673	0.0878	0.0007
	SimpleKT	dense	R	1,191,744	0.1756	0.2006	0.1563	0.0532	0.0085
		medium	R	129,494	0.2855	0.2994	0.1691	0.1349	0.0044
		sparse	R	10,079	0.1405	0.1491	0.1202	0.0391	0.0099
		Very Sparse	R	26,212	0.1002	0.1423	0.1404	0.0137	0.0109
		Strict Cold-start	R	233,214	0.2973	0.2896	0.1673	0.1223	0.0004

Note: Reliability flags (R/L/I) follow Section 3.5. These values serve as the numerical basis for Figure 3. Very Sparse excludes Strict Cold-start KCs; Strict Cold-start is reported separately where applicable. IRT calibration should be interpreted jointly with discrimination and Brier resolution: under learner-based splits, IRT behaves as a base-rate reference for unseen learners (Section 4.2); under temporal splits, it recovers non-zero resolution because learners are observed during training.

D Diagnostic Interpretation Guide

To assist researchers and practitioners in applying our diagnostic protocol, we provide Table 14 which maps common empirical diagnostic patterns observed during evaluation to their theoretical and operational educational interpretation.

Table 14: Diagnostic Interpretation Guide for Sparse-Concept and Calibration Analysis

Diagnostic Pattern	Interpretation / Recommendation
1. High overall AUC but high sparse-KC ECE	The model ranks responses well overall (good discrimination) but its predicted probabilities may be potentially unreliable (poor calibration) on low-frequency concepts. Educational interventions triggered by probability thresholds should be used with caution on these KCs.
2. Dense AUC remains stable but very-sparse AUC is unstable or has high variance (e.g., high standard deviations)	Check the number of KCs and interaction events in the stratum. Volatility in metrics can be driven by small sample-size constraints (metric instability) rather than intrinsic model shifts.
3. Brier reliability (REL) component increases in sparse buckets	Probabilistic calibration degrades under limited training evidence. Under our experimental conditions, deep sequence models tend to generate overconfident or under-confident probability predictions as concepts become scarcer.
4. Temporal split AUC converges near random guess (≈ 0.50), especially on strict cold-start groups	Future-oriented concept generalization can be challenging, especially under strict temporal cold-start constraints. However, if near-random AUC occurs on warm or dense cohorts where sufficient training evidence exists, the L8 predictive sanity check should be triggered to audit possible prediction-label alignment, sequence cut-off, export, or metric-computation errors before interpreting the result as genuine model failure.
5. strict, k5, and k10 cold-start groups show near-identical event counts or metrics	The dataset under study has very few genuinely cold-start concepts under the chosen split, or the KCs are densely linked, meaning the cold-start setting is not highly active for this cohort.
6. ECE and Brier score model rankings differ from AUC discrimination rankings	Discrimination and calibration capture orthogonal model characteristics. They answer different diagnostic questions and should be reported separately. Models optimized purely for pairwise ranking (AUC) may not produce reliable soft posterior masteries.

E Dataset Conditions for Sparse-KC Calibration Vulnerability

Table 15 accompanies Section 5.4. It reports only quantities that can be computed from training-fold frequencies, training-only KC covariates, and the event-level ECE already published in Table 12.

F When Sparse-Concept Diagnostics Are Informative

Tables 16 and 17 accompany Section 5.6. Need levels reuse the protocol’s pre-registered sparse threshold and reliability flags; C4 is qualitative intended use. The simulated gate in Section 4.5 assumes that High-C4 setting; it is not a measurement of C4 from the logs.

Table 15: Dataset conditions associated with observable sparse-KC calibration vulnerability. All six rows are structural or metric observations computed from training-fold frequencies and published event-level SimpleKT ECE (Table 12); they are FINDINGS, not causal claims. Hierarchical curriculum, ceiling effects, item-feature richness, and tagging granularity are *not* listed here because they are untested (HYPOTHESIS; see text). Sparse mass is the share of KCs with $f_{\text{train}} < 100$ (sparse + very sparse + strict). Test support uses the protocol reliability flags (R/L/I). Coupling is Spearman ρ between $\log(1 + f_{\text{train}})$ and the named training-only covariate.

Condition	ASSISTments 2012	Junyi Academy	XES3G5M
Meaningful sparse mass	Present (18.9% of KCs)	Absent (0.0% of KCs)	Present (22.5% of KCs)
Sufficient test support	Partial (sparse $N = 415$ L; very-sparse $N = 16$ I)	Absent (0 sparse/very-sparse test events)	Present (sparse $N = 2,010$ R; very-sparse $N = 112$ L)
Frequency–difficulty coupling	Present, expected dir. ($\rho = -0.227$; sparse harder)	Present, expected dir. ($\rho = -0.416$; sparse harder)	Weak, opposite-signed ($\rho = +0.087$)
Curriculum-position coupling	Present ($\rho = -0.308$)	Present ($\rho = -0.324$)	Weak ($\rho = -0.125$)
Item-support imbalance	High (dense 205.0 vs sparse 1.0 items/KC)	Median 18.0 items/KC (no sparse contrast)	High (dense 9.0 vs sparse 1.0 items/KC)
Observed calibration gradient (SimpleKT, event-level)	Monotonic increase (0.114 $\rightarrow$ 0.154 $\rightarrow$ 0.228)	Dense $\rightarrow$ medium only (empty) ($\Delta\text{ECE} = +0.028$); sparse empty	Flat dense $\rightarrow$ sparse (0.114 $\rightarrow$ 0.111 $\rightarrow$ 0.125)

Note: Event-level ECE matches Table 12. XES3G5M IRT ECE *decreases* from dense to sparse (inverted), whereas DKT increases; the SimpleKT row is therefore not a dataset-universal gradient. Junyi temporal splits do populate sparse strata (see Table 11), so the learner-based empty-sparse pattern is threshold- and volume-dependent.

G Controlled Sparsification of Training Evidence

The selection rule was recorded before reduced-evidence ECE was inspected (`controlled_sparsification_protocol.md`). On each dataset, eligible KCs had $f_{\text{train}} \geq 500$, $N_{\text{test}} \geq 100$, and at least 20 positive and 20 negative test labels. If more than 30 KCs qualified, ten were taken from each tertile of the training-only difficulty proxy $1 - \bar{c}_{\text{train}}$, sorted by `kc_id`. ECE was not a selection criterion.

Training rows of selected KCs were downsampled to 500, 100, or 50 interactions with a deterministic RNG; validation, test, KC identifiers, and non-selected training rows were unchanged. Official processed `train.csv` files were not overwritten. DKT and SimpleKT were retrained on ASSISTments 2012, Junyi Academy, and XES3G5M, learner-based fold 0, seed 42. The 100% control is the published prediction export for that seed, not a second A9 retrain of the full fold.

Primary outcomes are within-KC ΔECE and ΔREL (Table 18, Figure 4). A within-KC ECE increase after reducing training rows is not universal: it is absent on ASSISTments 2012, large for SimpleKT on Junyi Academy, and modest for SimpleKT on XES3G5M. We do not claim that real-world sparsity always causes miscalibration.

Table 16: Decision framework for when sparse-concept diagnostics are informative. Levels reuse the protocol’s pre-registered sparse threshold ($f_{\text{train}} < 100$) and reliability flags (R/L/I); they are not new cutoffs tuned on these three datasets. C4 is an intended-use condition, not a quantity estimated from the interaction logs. Section 4.5 simulates a global probability gate *assuming* C4 is High; it does not infer C4 from the logs. Reporting occupancy and reliability flags remains useful even when the empirical priority of sparse-stratum claims is Low.

Condition	Low need	Moderate need	High need
C1 Sparse mass	No KCs below the pre-registered sparse threshold ($f_{\text{train}} < 100$)	Sparse-like KCs appear only under an alternate split, or the tail is present but tiny relative to the dense mass	A non-empty sparse-like share of KCs under the split actually used (observed here: 18.9% and 22.5%)
C2 Evaluation support	Sparse strata empty or Insufficient ($N < 100$)	Sparse stratum Limited ($100 \leq N < 1000$)	Sparse stratum Reliable ($N \geq 1000$)
C3 Calibration susceptibility	No stratum-level ECE/REL rise and only a weak KC-level frequency–ECE association	KC-level association present, but the stratum gradient is flat, inverted, or model-heterogeneous	Monotonic ECE/REL increase from dense to sparse with at least Limited support
C4 Deployment relevance (qualitative)	Rankings only; probabilities are not thresholded	Probabilities are reported but not used as gates	Remediation, practice, or mastery decisions use probability thresholds
C5 Temporal / cold-start exposure	Temporal split does not populate strict or limited cold-start groups	Cold-start groups exist but are small in KC count	Large cold-start test mass (Reliable N) or worse-than-random discrimination on unseen KCs

Note: Low/Moderate/High describe the priority of *sparse-concept diagnostic claims*, not whether the protocol should be run. Occupancy reporting (empty buckets, reliability flags) is useful hygiene on every split we consider.

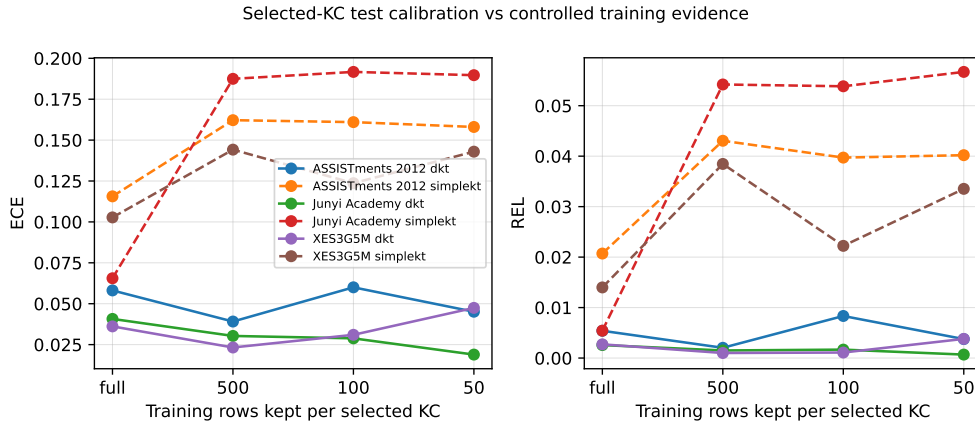


Figure 4: Selected-KC test ECE and REL versus training rows kept per selected KC on ASSISTments 2012, Junyi Academy, and XES3G5M (seed 42). The full points are the published models. This figure is descriptive; inference uses the paired within-KC bootstrap in Table 18.

H Bayesian Knowledge Tracing Numerical Instability and Transition

During our initial protocol implementation, we instantiated Bayesian Knowledge Tracing (BKT) as the classical baseline utilizing the popular `pyBKT` library (v1.4.1). However, on

Table 17: Sparse-concept diagnostic need on the three evaluation datasets. Need is not the same as observed vulnerability: XES3G5M has high reporting priority because sparse mass and cold-start support are large, even though SimpleKT ECE is flat. Junyi has low learner-based sparse-stratum priority because those buckets are empty, not because calibration is known to be safe. C4 is the same qualitative intended-use condition for all three public benchmarks.

Condition	ASSISTments 2012	Junyi Academy	XES3G5M
C1 Sparse mass	High: 18.9% of KCs below sparse threshold	Low: 0.0% of KCs with $f_{\text{train}} < 100$	High: 22.5% of KCs below sparse threshold
C2 Evaluation support	Moderate: sparse $N = 415$ (L); very-sparse $N = 16$ (I)	Low: 0 sparse/very-sparse test events	High: sparse $N = 2,010$ (R); very-sparse $N = 112$ (L)
C3 Calibration susceptibility	High: monotonic ECE $0.114 \rightarrow 0.154 \rightarrow 0.228$; $\Delta\text{ECE} = +0.028$; $\bar{\rho} = -0.60$	Moderate: dense $\rightarrow$ medium sparse empty; $\bar{\rho} = -0.43$	Moderate: SimpleKT flat $0.114 \rightarrow 0.125$; model-heterogeneous; $\bar{\rho} = -0.23$
C4 Deployment relevance	Use-dependent: not measured in logs; High if probabilities are thresholded	Use-dependent: not measured in logs; High if probabilities are thresholded	Use-dependent: not measured in logs; High if probabilities are thresholded
C5 Temporal / cold-start	Moderate: 27 strict KCs / 1,407 events; 85 temporal sparse-like KCs	Moderate: 4 strict KCs / 2,545 events; 10 temporal sparse-like KCs	High: 117 strict KCs / 233,214 events (R); 273 temporal sparse-like KCs
Overall sparse-diagnostic priority	High: C1+C3 high, C2 Limited but estimable; sparse ECE claims are informative	Low (learner-based); Moderate (temporal): empty sparse buckets under learner-based split; temporal split still exposes a small tail	High (reporting), Moderate (SimpleKT gradient): mass and support require the check; the check shows a flat/inverted pattern, not ASSISTments-like degradation

Note: Learner-based occupancy and ECE from Tables 15 and 12; temporal cold-start counts from Table 9. Overall priority is a qualitative synthesis, not a numeric score.

Table 18: Controlled sparsification: within-KC change in primary calibration outcomes on the held-out test events of 30 originally dense KCs per dataset (seed 42). Δ = reduced – published full. Positive values mean worse calibration after reducing training rows for the same KC. 95% CIs are bootstrap over KCs. This is a validation experiment; it does not claim that real-world sparsity always causes miscalibration.

Dataset	Model	Rows kept	ΔECE [95% CI]	ΔREL [95% CI]	Share ECE $\uparrow$
ASSISTments 2012	DKT	500	-0.047 [-0.060, -0.033]	-0.015 [-0.022, -0.010]	10%
ASSISTments 2012	DKT	100	-0.024 [-0.046, -0.001]	-0.008 [-0.017, 0.001]	30%
ASSISTments 2012	DKT	50	-0.017 [-0.034, 0.002]	-0.008 [-0.016, -0.002]	43%
ASSISTments 2012	SimpleKT	500	-0.026 [-0.047, -0.002]	-0.008 [-0.017, 0.003]	20%
ASSISTments 2012	SimpleKT	100	0.002 [-0.023, 0.029]	0.000 [-0.013, 0.014]	43%
ASSISTments 2012	SimpleKT	50	0.002 [-0.021, 0.025]	0.001 [-0.011, 0.012]	47%
Junyi Academy	DKT	500	-0.021 [-0.041, -0.001]	-0.007 [-0.013, -0.002]	30%
Junyi Academy	DKT	100	-0.010 [-0.031, 0.011]	-0.003 [-0.009, 0.002]	37%
Junyi Academy	DKT	50	-0.004 [-0.022, 0.013]	-0.002 [-0.008, 0.003]	57%
Junyi Academy	SimpleKT	500	0.101 [0.081, 0.120]	0.048 [0.035, 0.060]	93%
Junyi Academy	SimpleKT	100	0.132 [0.109, 0.155]	0.061 [0.047, 0.077]	100%
Junyi Academy	SimpleKT	50	0.135 [0.110, 0.161]	0.061 [0.047, 0.074]	100%
XES3G5M	DKT	500	-0.008 [-0.019, 0.004]	-0.002 [-0.005, 0.001]	27%
XES3G5M	DKT	100	0.001 [-0.009, 0.011]	-0.001 [-0.004, 0.002]	47%
XES3G5M	DKT	50	0.018 [0.006, 0.029]	0.003 [0.000, 0.006]	77%
XES3G5M	SimpleKT	500	0.014 [-0.002, 0.032]	0.011 [0.001, 0.023]	57%
XES3G5M	SimpleKT	100	0.015 [0.002, 0.030]	0.007 [0.001, 0.013]	63%
XES3G5M	SimpleKT	50	0.032 [0.021, 0.043]	0.018 [0.012, 0.025]	90%

all three large-scale datasets, BKT suffered from severe numerical instability during EM parameter estimation. Specifically, for sparse and very sparse concept strata, the scarcity of interaction logs triggered a divide-by-zero error in the M-step update equation for the prior parameter, resulting in NaN value assignments. Consequently, the model’s predictions degenerated into deterministic binary outputs ($p_{pred} \in \{0.0, \text{NaN}\}$), yielding a flat 0.50 AUC and extreme NLL values ranging between 23.7 and 28.0. To maintain scientific integrity and standard baseline comparison, we transitioned to Item Response Theory (IRT 1PL / Rasch Model) as the primary classical baseline. The raw prediction CSV files for the failed BKT runs are preserved within the codebase predictions directory for transparency.

I Prediction–Label Alignment Audit for Temporal Splits

During the temporal-split audit, we identified a prediction–label alignment issue in the temporal evaluation pipeline. The issue occurred when exported model predictions were matched back to temporal test instances after sequence construction and cohort filtering. In the affected code path, an alignment discrepancy in the temporal evaluation pipeline made some exported p_{pred} values correspond to incorrect y_{true} entries.

The main symptom was not limited to strict cold-start KCs. Deep KT baselines produced near-random AUC even on warm temporal cohorts, where KCs had sufficient training-fold evidence. This behavior was a red flag because a static IRT baseline still achieved meaningful AUC on the same warm cohorts, whereas DKT and SimpleKT collapsed to approximately random ranking.

We detected the issue using two sanity checks. First, we compared warm-cohort AUC against the IRT reference baseline; near-random deep KT AUC on warm KCs was treated as an alignment warning rather than as evidence of genuine temporal generalization failure. Second, we computed prediction–label correlation on warm-cohort slices and verified whether $\mathbb{E}[p_{pred} \mid y = 1]$ exceeded $\mathbb{E}[p_{pred} \mid y = 0]$. The correction preserves a stable test-instance identity through temporal splitting, sequence construction, prediction export, cohort filtering, and metric computation. Metrics are computed only after pairing predictions and labels by this stable identity rather than by transient row position.

After applying the correction consistently across datasets, warm-cohort AUC for the affected deep KT baselines recovered above the sanity threshold, and prediction–label correlations became positive. These checks were then used before re-running the temporal diagnostics reported in the paper. This audit reinforces the central message of the protocol: temporal cold-start results should be accompanied by warm-cohort and prediction–label sanity checks to distinguish genuine concept-level cold-start difficulty from alignment artifacts.

J Controlled Fault Injection for Predictive-Sanity Validation

The alignment case in Appendix I is one observed defect. A single catch does not by itself establish how L8 behaves under other alignment failures. This section reports a *validation experiment*—not a pre-registered primary analysis—that injects controlled faults into *in-memory copies* of the existing temporal prediction CSVs (DKT and SimpleKT, seed 42, three datasets). The original prediction files are never overwritten.

Fault classes. F1 and F2 pair p_t with y_{t+1} and y_{t-1} within each learner. F3 shifts p_{pred} by one timestep within learner. F4 shuffles predictions within learner. F5 reassigns p_{pred} by row index after a different sort (a global position mismatch of the class that motivated the original L8 catch). F6 swaps a random 1%, 5%, 10%, 25%, or 50% of rows.

Detection signals and rule. L8 records warm-cohort AUC, $\text{corr}(p_{\text{pred}}, y)$, $\mathbb{E}[p_{\text{pred}} | y = 1] - \mathbb{E}[p_{\text{pred}} | y = 0]$, the fraction of missing or unmatched stable ids, and prediction–label identity consistency against the official temporal test split. A file is flagged if any of the following hold: warm AUC < 0.55 ; warm AUC more than 0.05 below the clean IRT warm-cohort reference; non-positive correlation or conditional-mean gap; or an identity-mismatch rate more than 1 percentage point above that same file’s clean-pipeline identity-mismatch rate. The last clause uses the correct pipeline as the reference distribution: Junyi Academy has a small baseline identity collision ($\approx 0.9\%$) from duplicate stable ids, so an absolute zero-tolerance cutoff would false-positive the clean export. Thresholds were not chosen after inspecting the injected-fault outcomes.

Results. Table 19 reports detection rates over six files (two models $\times$ three datasets). The clean copies produced 0/6 false positives. L8 detected all F1, F2, and F5 injections (6/6 each). F1 and F2 were caught by the identity check: adjacent-label shifts left warm AUC in the 0.60–0.81 range because learner sequences are autocorrelated. F5—the row-index mismatch class—collapsed warm AUC to ≈ 0.50 and fired the original Appendix I signals. F3 was never detected (0/6): shifting p_{pred} by one step does not change exported labels or ids, and residual association remains positive. F4 was detected on 2/6 files (XES3G5M only, via collapse versus IRT). F6 was undetected at corruption rates $\leq 25\%$ and reached 3/6 at 50% (Figure 5). Overall detection was 23/60 injected faults (38.3%).

L8 is therefore sensitive to the identity-shifted-label and global row-index-mismatch classes tested here, and only weakly sensitive to within-learner shuffles and high-rate partial swaps. It is *not* sensitive to adjacent-timestep prediction shifts that preserve autocorrelation. We do not claim that L8 detects every evaluation bug.

Declaration of Generative AI Software Tools in the Writing Process

During the preparation of this work, the authors used ChatGPT, Claude and Google Antigravity to support language polishing, LaTeX formatting, consistency checking, and preparation of reproducibility prompts in the Abstract, Introduction, Methods, Results, Discussion, and Appendix sections. These tools were not used to generate, fabricate, or alter experimental

Table 19: Controlled fault-injection validation of channel L8 (temporal warm cohort). This is a validation experiment, not a pre-registered primary analysis. Detection uses the Appendix I signals plus identity consistency against official test-split labels on a stable instance id: warm AUC < 0.55 , or warm AUC more than 0.05 below the clean IRT warm-cohort reference, or non-positive $\text{corr}(p, y)$ / conditional-mean gap, or exported y elevated more than 1 percentage point above the same file’s clean-pipeline identity-mismatch rate. Thresholds were not tuned on the injected faults. L8 is sensitive to some of the tested alignment-fault classes; it is not claimed to detect all bugs.

Condition	Detection rate
Clean pipeline (false-positive check)	0/6
F1 label shift +1	100% (6/6)
F2 label shift -1	100% (6/6)
F3 prediction shift	0% (0/6)
F4 within-learner shuffle	33% (2/6)
F5 row-index mismatch	100% (6/6)
F6 partial mismatch 1%	0% (0/6)
F6 partial mismatch 5%	0% (0/6)
F6 partial mismatch 10%	0% (0/6)
F6 partial mismatch 25%	0% (0/6)
F6 partial mismatch 50%	50% (3/6)
All injected faults	23/60 (38.3%)

Note: Counts pool DKT and SimpleKT $\times$ three datasets (six files). Original prediction CSVs are not modified. Adjacent-timestep prediction shifts (F3) often retain warm AUC because learner sequences are autocorrelated; L8 as specified does not treat that residual association as a failure.

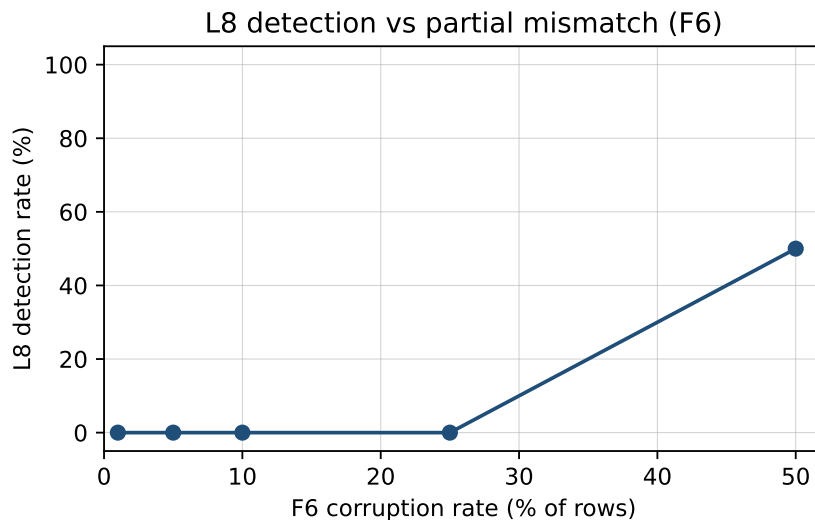


Figure 5: L8 detection rate versus F6 partial-mismatch corruption, pooled over six temporal prediction files. Detection remains 0 through 25% and reaches 50% (3/6 files) at 50% corruption. This is a validation curve, not a tuned operating characteristic.

results. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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